CHAPTER 2 Numeric Attributes

In this chapter, we discuss basic statistical methods for exploratory data analysis of numeric attributes. We look at measures of central tendency or location, measures of dispersion, and measures of linear dependence or association between attributes. We emphasize the connection between the probabilistic and the geometric and algebraic views of the data matrix.

2.1 UNIVARIATE ANALYSIS

Univariate analysis focuses on a single attribute at a time; thus the data matrix $\mathbf{D}$ can be thought of as an $n \times 1$ matrix, or simply a column vector, given as

$$\mathbf{D} = \begin{pmatrix} X \\ \hline x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

where X is the numeric attribute of interest, with $x_i \in \mathbb{R}$. X is assumed to be a random variable, with each point x_i $(1 \leq i \leq n)$ itself treated as an identity random variable. We assume that the observed data is a random sample drawn from X, that is, each variable x_i is independent and identically distributed as X. In the vector view, we treat the sample as an n-dimensional vector, and write $X \in \mathbb{R}^n$.

In general, the probability density or mass function $f(x)$ and the cumulative distribution function $F(x)$, for attribute X, are both unknown. However, we can estimate these distributions directly from the data sample, which also allow us to compute statistics to estimate several important population parameters.

Empirical Cumulative Distribution Function

The *empirical cumulative distribution function (CDF)* of X is given as

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} I(x_i \leq x) \tag{2.1}$$

where

$$I(x_i \le x) = \begin{cases} 1 & \text{if } x_i \le x \\ 0 & \text{if } x_i > x \end{cases}$$

is a binary *indicator variable* that indicates whether the given condition is satisfied or not. Intuitively, to obtain the empirical CDF we compute, for each value $x \in \mathbb{R}$, how many points in the sample are less than or equal to x. The empirical CDF puts a probability mass of $\frac{1}{n}$ at each point x_i. Note that we use the notation $\hat{F}$ to denote the fact that the empirical CDF is an estimate for the unknown population CDF F.

Inverse Cumulative Distribution Function

Define the *inverse cumulative distribution function* or *quantile function* for a random variable X as follows:

$$F^{-1}(q) = \min\{x \mid F(x) \ge q\} \qquad \text{for } q \in [0,1] \tag{2.2}$$

That is, the inverse CDF gives the least value of X, for which q fraction of the values are lower, and $1-q$ fraction of the values are higher. The *empirical inverse cumulative distribution function* $\hat{F}^{-1}$ can be obtained from Eq. (2.1).

Empirical Probability Mass Function

The *empirical probability mass function (PMF)* of X is given as

$$\hat{f}(x) = P(X = x) = \frac{1}{n}\sum_{i=1}^{n} I(x_i = x) \tag{2.3}$$

where

$$I(x_i = x) = \begin{cases} 1 & \text{if } x_i = x \\ 0 & \text{if } x_i \ne x \end{cases}$$

The empirical PMF also puts a probability mass of $\frac{1}{n}$ at each point x_i.

2.1.1 Measures of Central Tendency

These measures given an indication about the concentration of the probability mass, the "middle" values, and so on.

Mean

The *mean*, also called the *expected value*, of a random variable X is the arithmetic average of the values of X. It provides a one-number summary of the *location* or *central tendency* for the distribution of X.

The mean or expected value of a discrete random variable X is defined as

$$\mu = E[X] = \sum_{x} x f(x) \tag{2.4}$$

where $f(x)$ is the probability mass function of X.

The expected value of a continuous random variable X is defined as

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

where $f(x)$ is the probability density function of X.

Sample Mean The *sample mean* is a statistic, that is, a function $\hat{\mu} : \{x_1, x_2, \ldots, x_n\} \to \mathbb{R}$, defined as the average value of x_i's:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{2.5}$$

It serves as an estimator for the unknown mean value μ of X. It can be derived by plugging in the empirical PMF $\hat{f}(x)$ in Eq. (2.4):

$$\hat{\mu} = \sum_x x \hat{f}(x) = \sum_x x \left(\frac{1}{n} \sum_{i=1}^{n} I(x_i = x) \right) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Sample Mean Is Unbiased An estimator $\hat{\theta}$ is called an *unbiased estimator* for parameter θ if $E[\hat{\theta}] = \theta$ for every possible value of θ. The sample mean $\hat{\mu}$ is an unbiased estimator for the population mean μ, as

$$E[\hat{\mu}] = E\left[\frac{1}{n} \sum_{i=1}^{n} x_i \right] = \frac{1}{n} \sum_{i=1}^{n} E[x_i] = \frac{1}{n} \sum_{i=1}^{n} \mu = \mu \tag{2.6}$$

where we use the fact that the random variables x_i are IID according to X, which implies that they have the same mean μ as X, that is, $E[x_i] = \mu$ for all x_i. We also used the fact that the expectation function E is a *linear operator*, that is, for any two random variables X and Y, and real numbers a and b, we have $E[aX + bY] = aE[X] + bE[Y]$.

Robustness We say that a statistic is *robust* if it is not affected by extreme values (such as outliers) in the data. The sample mean is unfortunately not robust because a single large value (an outlier) can skew the average. A more robust measure is the *trimmed mean* obtained after discarding a small fraction of extreme values on one or both ends. Furthermore, the mean can be somewhat misleading in that it is typically not a value that occurs in the sample, and it may not even be a value that the random variable can actually assume (for a discrete random variable). For example, the number of cars per capita is an integer-valued random variable, but according to the US Bureau of Transportation Studies, the average number of passenger cars in the United States was 0.45 in 2008 (137.1 million cars, with a population size of 304.4 million). Obviously, one cannot own 0.45 cars; it can be interpreted as saying that on average there are 45 cars per 100 people.

Median

The *median* of a random variable is defined as the value m such that

$$P(X \le m) \ge \frac{1}{2} \text{ and } P(X \ge m) \ge \frac{1}{2}$$

In other words, the median m is the "middle-most" value; half of the values of X are less and half of the values of X are more than m. In terms of the (inverse) cumulative distribution function, the median is therefore the value m for which

$$F(m) = 0.5 \text{ or } m = F^{-1}(0.5)$$

The *sample median* can be obtained from the empirical CDF [Eq. (2.1)] or the empirical inverse CDF [Eq. (2.2)] by computing

$$\hat{F}(m) = 0.5 \text{ or } m = \hat{F}^{-1}(0.5)$$

A simpler approach to compute the sample median is to first sort all the values x_i ($i \in [1,n]$) in increasing order. If n is odd, the median is the value at position $\frac{n+1}{2}$. If n is even, the values at positions $\frac{n}{2}$ and $\frac{n}{2}+1$ are both medians.

Unlike the mean, median is robust, as it is not affected very much by extreme values. Also, it is a value that occurs in the sample and a value the random variable can actually assume.

Mode

The *mode* of a random variable X is the value at which the probability mass function or the probability density function attains its maximum value, depending on whether X is discrete or continuous, respectively.

The *sample mode* is a value for which the empirical probability mass function [Eq. (2.3)] attains its maximum, given as

$$\text{mode}(X) = \arg\max_x \hat{f}(x)$$

The mode may not be a very useful measure of central tendency for a sample because by chance an unrepresentative element may be the most frequent element. Furthermore, if all values in the sample are distinct, each of them will be the mode.

Example 2.1 (Sample Mean, Median, and Mode). Consider the attribute `sepal length` (X_1) in the Iris dataset, whose values are shown in Table 1.2. The sample mean is given as follows:

$$\hat{\mu} = \frac{1}{150}(5.9 + 6.9 + \cdots + 7.7 + 5.1) = \frac{876.5}{150} = 5.843$$

Figure 2.1 shows all 150 values of `sepal length`, and the sample mean. Figure 2.2a shows the empirical CDF and Figure 2.2b shows the empirical inverse CDF for `sepal length`.

Because $n = 150$ is even, the sample median is the value at positions $\frac{n}{2} = 75$ and $\frac{n}{2} + 1 = 76$ in sorted order. For `sepal length` both these values are 5.8; thus the sample median is 5.8. From the inverse CDF in Figure 2.2b, we can see that

$$\hat{F}(5.8) = 0.5 \text{ or } 5.8 = \hat{F}^{-1}(0.5)$$

The sample mode for `sepal length` is 5, which can be observed from the frequency of 5 in Figure 2.1. The empirical probability mass at $x = 5$ is

$$\hat{f}(5) = \frac{10}{150} = 0.067$$

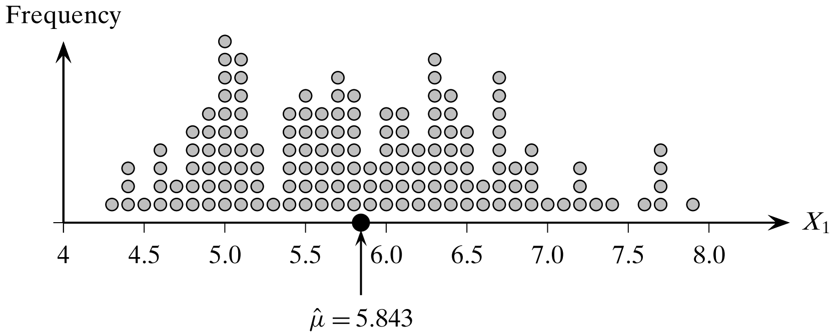

Figure 2.1. Sample mean for `sepal length`. Multiple occurrences of the same value are shown stacked.

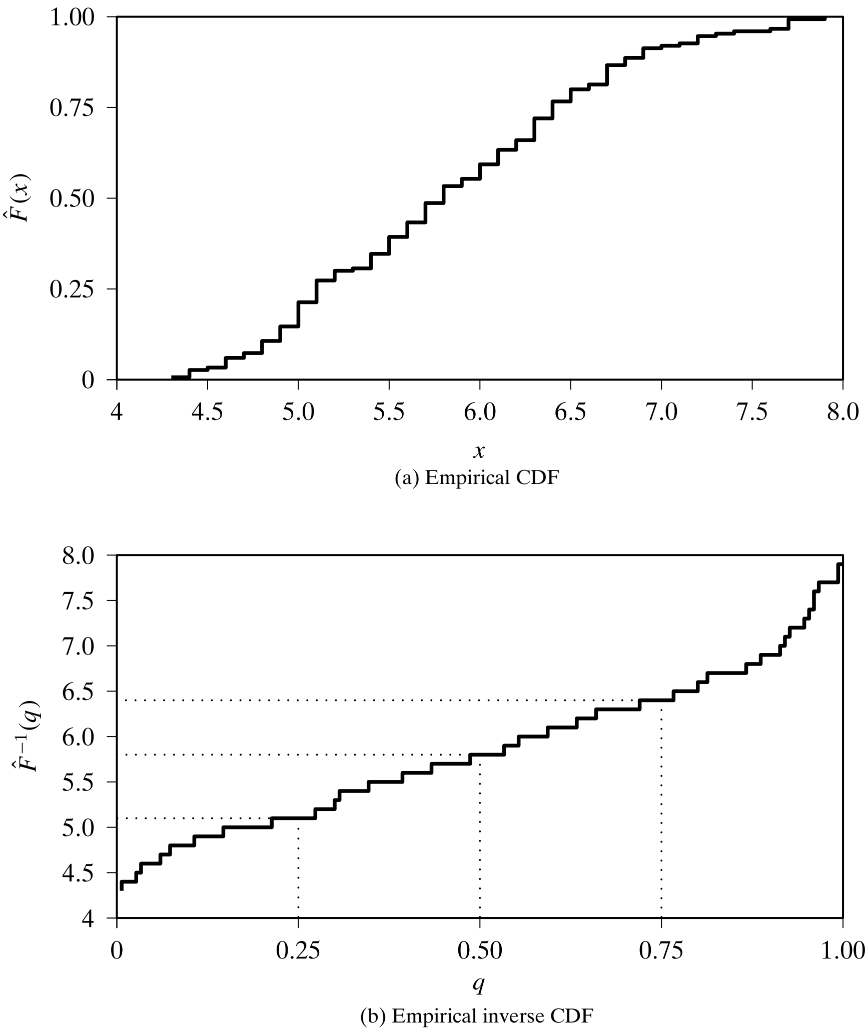

Figure 2.2. Empirical CDF and inverse CDF: `sepal length`.

2.1.2 Measures of Dispersion

The measures of dispersion give an indication about the spread or variation in the values of a random variable.

Range

The *value range* or simply *range* of a random variable X is the difference between the maximum and minimum values of X, given as

$$r = \max\{X\} - \min\{X\}$$

The (value) range of X is a population parameter, not to be confused with the range of the function X, which is the set of all the values X can assume. Which range is being used should be clear from the context.

The *sample range* is a statistic, given as

$$\hat{r} = \max_{i=1}^{n}\{x_i\} - \min_{i=1}^{n}\{x_i\}$$

By definition, range is sensitive to extreme values, and thus is not robust.

Interquartile Range

Quartiles are special values of the quantile function [Eq. (2.2)] that divide the data into four equal parts. That is, quartiles correspond to the quantile values of 0.25, 0.5, 0.75, and 1.0. The *first quartile* is the value $q_1 = F^{-1}(0.25)$, to the left of which 25% of the points lie; the *second quartile* is the same as the median value $q_2 = F^{-1}(0.5)$, to the left of which 50% of the points lie; the third quartile $q_3 = F^{-1}(0.75)$ is the value to the left of which 75% of the points lie; and the fourth quartile is the maximum value of X, to the left of which 100% of the points lie.

A more robust measure of the dispersion of X is the *interquartile range (IQR)*, defined as

$$IQR = q_3 - q_1 = F^{-1}(0.75) - F^{-1}(0.25) \tag{2.7}$$

IQR can also be thought of as a *trimmed range*, where we discard 25% of the low and high values of X. Or put differently, it is the range for the middle 50% of the values of X. IQR is robust by definition.

The *sample IQR* can be obtained by plugging in the empirical inverse CDF in Eq. (2.7):

$$\widehat{IQR} = \hat{q}_3 - \hat{q}_1 = \hat{F}^{-1}(0.75) - \hat{F}^{-1}(0.25)$$

Variance and Standard Deviation

The *variance* of a random variable X provides a measure of how much the values of X deviate from the mean or expected value of X. More formally, variance is the expected value of the squared deviation from the mean, defined as

$$\sigma^2 = var(X) = E[(X-\mu)^2] = \begin{cases} \sum_x (x-\mu)^2 f(x) & \text{if } X \text{ is discrete} \\ \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx & \text{if } X \text{ is continuous} \end{cases} \tag{2.8}$$

The *standard deviation*, σ, is defined as the positive square root of the variance, σ^2.

We can also write the variance as the difference between the expectation of X^2 and the square of the expectation of X:

$$\begin{aligned} \sigma^2 = var(X) &= E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - (E[X])^2 \end{aligned} \tag{2.9}$$

It is worth noting that variance is in fact the *second moment about the mean*, corresponding to $r = 2$, which is a special case of the *rth moment about the mean* for a random variable X, defined as $E[(\mathbf{x}-\mu)^r]$.

Sample Variance The *sample variance* is defined as

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2 \tag{2.10}$$

It is the average squared deviation of the data values x_i from the sample mean $\hat{\mu}$, and can be derived by plugging in the empirical probability function $\hat{f}$ from Eq. (2.3) into Eq. (2.8), as

$$\hat{\sigma}^2 = \sum_x (x-\hat{\mu})^2 \hat{f}(x) = \sum_x (x-\hat{\mu})^2 \left(\frac{1}{n}\sum_{i=1}^{n} I(x_i = x)\right) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2$$

The *sample standard deviation* is given as the positive square root of the sample variance:

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2}$$

The *standard score*, also called the *z-score*, of a sample value x_i is the number of standard deviations the value is away from the mean:

$$z_i = \frac{x_i - \hat{\mu}}{\hat{\sigma}}$$

Put differently, the z-score of x_i measures the deviation of x_i from the mean value $\hat{\mu}$, in units of $\hat{\sigma}$.

Geometric Interpretation of Sample Variance We can treat the data sample for attribute X as a vector in n-dimensional space, where n is the sample size. That is, we write $X = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$. Further, let

$$Z = X - \mathbf{1} \cdot \hat{\mu} = \begin{pmatrix} x_1 - \hat{\mu} \\ x_2 - \hat{\mu} \\ \vdots \\ x_n - \hat{\mu} \end{pmatrix}$$

denote the mean subtracted attribute vector, where $\mathbf{1} \in \mathbb{R}^n$ is the n-dimensional vector all of whose elements have value 1. We can rewrite Eq. (2.10) in terms of the magnitude of Z, that is, the dot product of Z with itself:

$$\hat{\sigma}^2 = \frac{1}{n} \|Z\|^2 = \frac{1}{n} Z^T Z = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2 \tag{2.11}$$

The sample variance can thus be interpreted as the squared magnitude of the centered attribute vector, or the dot product of the centered attribute vector with itself, normalized by the sample size.

Example 2.2. Consider the data sample for `sepal length` shown in Figure 2.1. We can see that the sample range is given as

$$\max_i\{x_i\} - \min_i\{x_i\} = 7.9 - 4.3 = 3.6$$

From the inverse CDF for `sepal length` in Figure 2.2b, we can find the sample IQR as follows:

$$\hat{q}_1 = \hat{F}^{-1}(0.25) = 5.1$$
$$\hat{q}_3 = \hat{F}^{-1}(0.75) = 6.4$$
$$\widehat{IQR} = \hat{q}_3 - \hat{q}_1 = 6.4 - 5.1 = 1.3$$

The sample variance can be computed from the centered data vector via Eq. (2.11):

$$\hat{\sigma}^2 = \frac{1}{n}(X - \mathbf{1} \cdot \hat{\mu})^T (X - \mathbf{1} \cdot \hat{\mu}) = 102.168/150 = 0.681$$

The sample standard deviation is then

$$\hat{\sigma} = \sqrt{0.681} = 0.825$$

Variance of the Sample Mean Because the sample mean $\hat{\mu}$ is itself a statistic, we can compute its mean value and variance. The expected value of the sample mean is simply μ, as we saw in Eq. (2.6). To derive an expression for the variance of the sample mean, we utilize the fact that the random variables x_i are all independent, and thus

$$var\left(\sum_{i=1}^{n} x_i\right) = \sum_{i=1}^{n} var(x_i)$$

Further, because all the x_i's are identically distributed as X, they have the same variance as X, that is,

$$var(x_i) = \sigma^2 \text{ for all } i$$

Combining the above two facts, we get

$$var\left(\sum_{i=1}^{n} x_i\right) = \sum_{i=1}^{n} var(x_i) = \sum_{i=1}^{n} \sigma^2 = n\sigma^2 \tag{2.12}$$

Further, note that

$$E\left[\sum_{i=1}^{n} x_i\right] = n\mu \tag{2.13}$$

Using Eqs. (2.9), (2.12), and (2.13), the variance of the sample mean $\hat{\mu}$ can be computed as

$$\begin{aligned}
var(\hat{\mu}) &= E[(\hat{\mu}-\mu)^2] = E[\hat{\mu}^2] - \mu^2 = E\left[\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)^2\right] - \frac{1}{n^2}E\left[\sum_{i=1}^{n} x_i\right]^2 \\
&= \frac{1}{n^2}\left(E\left[\left(\sum_{i=1}^{n} x_i\right)^2\right] - E\left[\sum_{i=1}^{n} x_i\right]^2\right) = \frac{1}{n^2}\, var\left(\sum_{i=1}^{n} x_i\right) \\
&= \frac{\sigma^2}{n}
\end{aligned} \tag{2.14}$$

In other words, the sample mean $\hat{\mu}$ varies or deviates from the mean μ in proportion to the population variance σ^2. However, the deviation can be made smaller by considering larger sample size n.

Sample Variance Is Biased, but Is Asymptotically Unbiased The sample variance in Eq. (2.10) is a *biased estimator* for the true population variance, σ^2, that is, $E[\hat{\sigma}^2] \neq \sigma^2$. To show this we make use of the identity

$$\sum_{i=1}^{n}(x_i-\mu)^2 = n(\hat{\mu}-\mu)^2 + \sum_{i=1}^{n}(x_i-\hat{\mu})^2 \tag{2.15}$$

Computing the expectation of $\hat{\sigma}^2$ by using Eq. (2.15) in the first step, we get

$$E[\hat{\sigma}^2] = E\left[\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2\right] - E[(\hat{\mu}-\mu)^2] \tag{2.16}$$

Recall that the random variables x_i are IID according to X, which means that they have the same mean μ and variance σ^2 as X. This means that

$$E[(x_i - \mu)^2] = \sigma^2$$

Further, from Eq. (2.14) the sample mean $\hat{\mu}$ has variance $E[(\hat{\mu} - \mu)^2] = \frac{\sigma^2}{n}$. Plugging these into the Eq. (2.16) we get

$$\begin{aligned} E[\hat{\sigma}^2] &= \frac{1}{n} n\sigma^2 - \frac{\sigma^2}{n} \\ &= \left(\frac{n-1}{n}\right)\sigma^2 \end{aligned}$$

The sample variance $\hat{\sigma}^2$ is a biased estimator of σ^2, as its expected value differs from the population variance by a factor of $\frac{n-1}{n}$. However, it is *asymptotically unbiased*, that is, the bias vanishes as $n \to \infty$ because

$$\lim_{n\to\infty} \frac{n-1}{n} = \lim_{n\to\infty} 1 - \frac{1}{n} = 1$$

Put differently, as the sample size increases, we have

$$E[\hat{\sigma}^2] \to \sigma^2 \qquad \text{as } n \to \infty$$

2.2 BIVARIATE ANALYSIS

In bivariate analysis, we consider two attributes at the same time. We are specifically interested in understanding the association or dependence between them, if any. We thus restrict our attention to the two numeric attributes of interest, say X_1 and X_2, with the data $\mathbf{D}$ represented as an $n \times 2$ matrix:

$$\mathbf{D} = \begin{pmatrix} X_1 & X_2 \\ \hline x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}$$

Geometrically, we can think of $\mathbf{D}$ in two ways. It can be viewed as n points or vectors in 2-dimensional space over the attributes X_1 and X_2, that is, $\mathbf{x}_i = (x_{i1}, x_{i2})^T \in \mathbb{R}^2$. Alternatively, it can be viewed as two points or vectors in an n-dimensional space comprising the points, that is, each column is a vector in $\mathbb{R}^n$, as follows:

$$\begin{aligned} X_1 &= (x_{11}, x_{21}, \ldots, x_{n1})^T \\ X_2 &= (x_{12}, x_{22}, \ldots, x_{n2})^T \end{aligned}$$

In the probabilistic view, the column vector $\mathbf{X} = (X_1, X_2)^T$ is considered a bivariate vector random variable, and the points $\mathbf{x}_i$ $(1 \le i \le n)$ are treated as a random sample drawn from $\mathbf{X}$, that is, $\mathbf{x}_i$'s are considered independent and identically distributed as $\mathbf{X}$.

Empirical Joint Probability Mass Function

The *empirical joint probability mass function* for $\mathbf{X}$ is given as

$$\hat{f}(\mathbf{x}) = P(\mathbf{X} = \mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} I(\mathbf{x}_i = \mathbf{x}) \tag{2.17}$$

$$\hat{f}(x_1, x_2) = P(X_1 = x_1, X_2 = x_2) = \frac{1}{n} \sum_{i=1}^{n} I(x_{i1} = x_1, x_{i2} = x_2)$$

where $\mathbf{x} = (x_1, x_2)^T$ and I is a indicator variable that takes on the value 1 only when its argument is true:

$$I(\mathbf{x}_i = \mathbf{x}) = \begin{cases} 1 & \text{if } x_{i1} = x_1 \text{ and } x_{i2} = x_2 \\ 0 & \text{otherwise} \end{cases}$$

As in the univariate case, the probability function puts a probability mass of $\frac{1}{n}$ at each point in the data sample.

2.2.1 Measures of Location and Dispersion

Mean

The bivariate mean is defined as the expected value of the vector random variable $\mathbf{X}$, defined as follows:

$$\boldsymbol{\mu} = E[\mathbf{X}] = E\left[\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}\right] = \begin{pmatrix} E[X_1] \\ E[X_2] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \tag{2.18}$$

In other words, the bivariate mean vector is simply the vector of expected values along each attribute.

The sample mean vector can be obtained from $\hat{f}_{X_1}$ and $\hat{f}_{X_2}$, the empirical probability mass functions of X_1 and X_2, respectively, using Eq. (2.5). It can also be computed from the joint empirical PMF in Eq. (2.17)

$$\hat{\boldsymbol{\mu}} = \sum_{\mathbf{x}} \mathbf{x} \hat{f}(\mathbf{x}) = \sum_{\mathbf{x}} \mathbf{x} \left(\frac{1}{n} \sum_{i=1}^{n} I(\mathbf{x}_i = \mathbf{x}) \right) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \tag{2.19}$$

Variance

We can compute the variance along each attribute, namely σ_1^2 for X_1 and σ_2^2 for X_2 using Eq. (2.8). The *total variance* [Eq. (1.4)] is given as

$$var(\mathbf{D}) = \sigma_1^2 + \sigma_2^2$$

The sample variances $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$ can be estimated using Eq. (2.10), and the *sample total variance* is simply $\hat{\sigma}_1^2 + \hat{\sigma}_2^2$.

2.2.2 Measures of Association

Covariance

The *covariance* between two attributes X_1 and X_2 provides a measure of the association or linear dependence between them, and is defined as

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] \tag{2.20}$$

By linearity of expectation, we have

$$\begin{aligned}
\sigma_{12} &= E[(X_1 - \mu_1)(X_2 - \mu_2)] \\
&= E[X_1X_2 - X_1\mu_2 - X_2\mu_1 + \mu_1\mu_2] \\
&= E[X_1X_2] - \mu_2 E[X_1] - \mu_1 E[X_2] + \mu_1\mu_2 \\
&= E[X_1X_2] - \mu_1\mu_2 \\
&= E[X_1X_2] - E[X_1]E[X_2] \qquad (2.21)
\end{aligned}$$

Eq. (2.21) can be seen as a generalization of the univariate variance [Eq. (2.9)] to the bivariate case.

If X_1 and X_2 are independent random variables, then we conclude that their covariance is zero. This is because if X_1 and X_2 are independent, then we have

$$E[X_1X_2] = E[X_1] \cdot E[X_2]$$

which in turn implies that

$$\sigma_{12} = 0$$

However, the converse is not true. That is, if $\sigma_{12} = 0$, one cannot claim that X_1 and X_2 are independent. All we can say is that there is no linear dependence between them, but we cannot rule out that there might be a higher order relationship or dependence between the two attributes.

The *sample covariance* between X_1 and X_2 is given as

$$\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^{n} (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2) \qquad (2.22)$$

It can be derived by substituting the empirical joint probability mass function $\hat{f}(x_1, x_2)$ from Eq. (2.17) into Eq. (2.20), as follows:

$$\begin{aligned}
\hat{\sigma}_{12} &= E[(X_1 - \hat{\mu}_1)(X_2 - \hat{\mu}_2)] \\
&= \sum_{\mathbf{x}=(x_1,x_2)^T} (x_1 - \hat{\mu}_1)(x_2 - \hat{\mu}_2)\hat{f}(x_1, x_2) \\
&= \frac{1}{n} \sum_{\mathbf{x}=(x_1,x_2)^T} \sum_{i=1}^{n} (x_1 - \hat{\mu}_1) \cdot (x_2 - \hat{\mu}_2) \cdot I(x_{i1} = x_1, x_{i2} = x_2) \\
&= \frac{1}{n} \sum_{i=1}^{n} (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)
\end{aligned}$$

Notice that sample covariance is a generalization of the sample variance [Eq. (2.10)] because

$$\hat{\sigma}_{11} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_1)(x_i - \mu_1) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_1)^2 = \hat{\sigma}_1^2$$

and similarly, $\hat{\sigma}_{22} = \hat{\sigma}_2^2$.

Correlation

The *correlation* between variables X_1 and X_2 is the *standardized covariance*, obtained by normalizing the covariance with the standard deviation of each variable, given as

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}} \tag{2.23}$$

The *sample correlation* for attributes X_1 and X_2 is given as

$$\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^n (x_{i2} - \hat{\mu}_2)^2}} \tag{2.24}$$

Geometric Interpretation of Sample Covariance and Correlation

Let Z_1 and Z_2 denote the centered attribute vectors in $\mathbb{R}^n$, given as follows:

$$Z_1 = X_1 - \mathbf{1} \cdot \hat{\mu}_1 = \begin{pmatrix} x_{11} - \hat{\mu}_1 \\ x_{21} - \hat{\mu}_1 \\ \vdots \\ x_{n1} - \hat{\mu}_1 \end{pmatrix} \qquad Z_2 = X_2 - \mathbf{1} \cdot \hat{\mu}_2 = \begin{pmatrix} x_{12} - \hat{\mu}_2 \\ x_{22} - \hat{\mu}_2 \\ \vdots \\ x_{n2} - \hat{\mu}_2 \end{pmatrix}$$

The sample covariance [Eq. (2.22)] can then be written as

$$\hat{\sigma}_{12} = \frac{Z_1^T Z_2}{n}$$

In other words, the covariance between the two attributes is simply the dot product between the two centered attribute vectors, normalized by the sample size. The above can be seen as a generalization of the univariate sample variance given in Eq. (2.11).

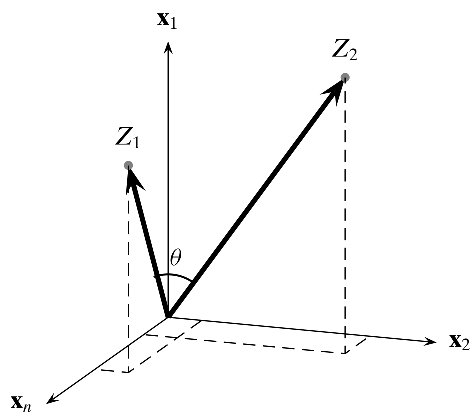

Figure 2.3. Geometric interpretation of covariance and correlation. The two centered attribute vectors are shown in the (conceptual) n-dimensional space $\mathbb{R}^n$ spanned by the n points.

The sample correlation [Eq. (2.24)] can be written as

$$\hat{\rho}_{12} = \frac{Z_1^T Z_2}{\sqrt{Z_1^T Z_1}\sqrt{Z_2^T Z_2}} = \frac{Z_1^T Z_2}{\|Z_1\|\ \|Z_2\|} = \left(\frac{Z_1}{\|Z_1\|}\right)^T \left(\frac{Z_2}{\|Z_2\|}\right) = \cos\theta \tag{2.25}$$

Thus, the correlation coefficient is simply the cosine of the angle [Eq. (1.3)] between the two centered attribute vectors, as illustrated in Figure 2.3.

Covariance Matrix

The variance–covariance information for the two attributes X_1 and X_2 can be summarized in the square 2×2 *covariance matrix*, given as

$$\begin{aligned}
\boldsymbol{\Sigma} &= E[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^T] \\
&= E\left[\begin{pmatrix} X_1-\mu_1 \\ X_2-\mu_2 \end{pmatrix}\begin{pmatrix} X_1-\mu_1 & X_2-\mu_2 \end{pmatrix}\right] \\
&= \begin{pmatrix} E[(X_1-\mu_1)(X_1-\mu_1)] & E[(X_1-\mu_1)(X_2-\mu_2)] \\ E[(X_2-\mu_2)(X_1-\mu_1)] & E[(X_2-\mu_2)(X_2-\mu_2)] \end{pmatrix} \\
&= \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}
\end{aligned} \tag{2.26}$$

Because $\sigma_{12} = \sigma_{21}$, $\boldsymbol{\Sigma}$ is a *symmetric* matrix. The covariance matrix records the attribute specific variances on the main diagonal, and the covariance information on the off-diagonal elements.

The *total variance* of the two attributes is given as the sum of the diagonal elements of $\boldsymbol{\Sigma}$, which is also called the *trace* of $\boldsymbol{\Sigma}$, given as

$$var(\mathbf{D}) = tr(\boldsymbol{\Sigma}) = \sigma_1^2 + \sigma_2^2$$

We immediately have $tr(\boldsymbol{\Sigma}) \geq 0$.

The *generalized variance* of the two attributes also considers the covariance, in addition to the attribute variances, and is given as the *determinant* of the covariance matrix $\boldsymbol{\Sigma}$, denoted as $|\boldsymbol{\Sigma}|$ or $\det(\boldsymbol{\Sigma})$. The generalized covariance is non-negative, because

$$|\boldsymbol{\Sigma}| = \det(\boldsymbol{\Sigma}) = \sigma_1^2\sigma_2^2 - \sigma_{12}^2 = \sigma_1^2\sigma_2^2 - \rho_{12}^2\sigma_1^2\sigma_2^2 = (1-\rho_{12}^2)\sigma_1^2\sigma_2^2$$

where we used Eq. (2.23), that is, $\sigma_{12} = \rho_{12}\sigma_1\sigma_2$. Note that $|\rho_{12}| \leq 1$ implies that $\rho_{12}^2 \leq 1$, which in turn implies that $\det(\boldsymbol{\Sigma}) \geq 0$, that is, the determinant is non-negative.

The *sample covariance matrix* is given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 \end{pmatrix}$$

The sample covariance matrix $\widehat{\boldsymbol{\Sigma}}$ shares the same properties as $\boldsymbol{\Sigma}$, that is, it is symmetric and $|\widehat{\boldsymbol{\Sigma}}| \geq 0$, and it can be used to easily obtain the sample total and generalized variance.

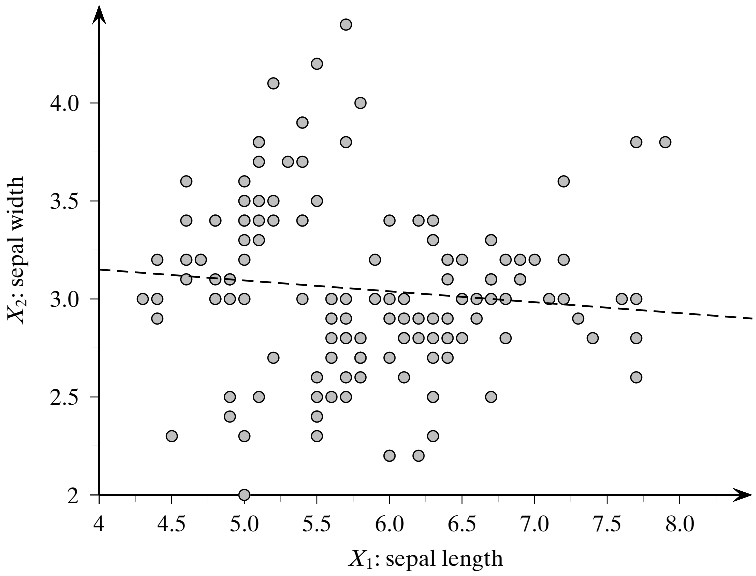

Figure 2.4. Correlation between `sepal length` and `sepal width`.

Example 2.3 (Sample Mean and Covariance). Consider the `sepal length` and `sepal width` attributes for the Iris dataset, plotted in Figure 2.4. There are $n = 150$ points in the $d = 2$ dimensional attribute space. The sample mean vector is given as

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} 5.843 \\ 3.054 \end{pmatrix}$$

The sample covariance matrix is given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} 0.681 & -0.039 \\ -0.039 & 0.187 \end{pmatrix}$$

The variance for `sepal length` is $\hat{\sigma}_1^2 = 0.681$, and that for `sepal width` is $\hat{\sigma}_2^2 = 0.187$. The covariance between the two attributes is $\hat{\sigma}_{12} = -0.039$, and the correlation between them is

$$\hat{\rho}_{12} = \frac{-0.039}{\sqrt{0.681 \cdot 0.187}} = -0.109$$

Thus, there is a very weak negative correlation between these two attributes, as evidenced by the best linear fit line in Figure 2.4. Alternatively, we can consider the attributes `sepal length` and `sepal width` as two points in $\mathbb{R}^n$. The correlation is then the cosine of the angle between them; we have

$$\hat{\rho}_{12} = \cos\theta = -0.109, \text{ which implies that } \theta = \cos^{-1}(-0.109) = 96.26^\circ$$

The angle is close to 90°, that is, the two attribute vectors are almost orthogonal, indicating weak correlation. Further, the angle being greater than 90° indicates negative correlation.

The sample total variance is given as

$$tr(\widehat{\mathbf{\Sigma}}) = 0.681 + 0.187 = 0.868$$

and the sample generalized variance is given as

$$|\widehat{\mathbf{\Sigma}}| = \det(\widehat{\mathbf{\Sigma}}) = 0.681 \cdot 0.187 - (-0.039)^2 = 0.126$$

2.3 MULTIVARIATE ANALYSIS

In multivariate analysis, we consider all the d numeric attributes $X_1, X_2, \ldots, X_d$. The full data is an $n \times d$ matrix, given as

$$\mathbf{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_d \\ \hline x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}$$

In the row view, the data can be considered as a set of n points or vectors in the d-dimensional attribute space

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{id})^T \in \mathbb{R}^d$$

In the column view, the data can be considered as a set of d points or vectors in the n-dimensional space spanned by the data points

$$X_j = (x_{1j}, x_{2j}, \ldots, x_{nj})^T \in \mathbb{R}^n$$

In the probabilistic view, the d attributes are modeled as a vector random variable, $\mathbf{X} = (X_1, X_2, \ldots, X_d)^T$, and the points $\mathbf{x}_i$ are considered to be a random sample drawn from $\mathbf{X}$, that is, they are independent and identically distributed as $\mathbf{X}$.

Mean

Generalizing Eq. (2.18), the *multivariate mean vector* is obtained by taking the mean of each attribute, given as

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{pmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_d] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_d \end{pmatrix}$$

Generalizing Eq. (2.19), the *sample mean* is given as

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i$$

Covariance Matrix

Generalizing Eq. (2.26) to d dimensions, the multivariate covariance information is captured by the $d \times d$ (square) symmetric *covariance matrix* that gives the covariance for each pair of attributes:

$$\boldsymbol{\Sigma} = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T] = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \cdots & \cdots & \cdots & \cdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{pmatrix}$$

The diagonal element σ_i^2 specifies the attribute variance for X_i, whereas the off-diagonal elements $\sigma_{ij} = \sigma_{ji}$ represent the covariance between attribute pairs X_i and X_j.

Covariance Matrix Is Positive Semidefinite

It is worth noting that $\boldsymbol{\Sigma}$ is a *positive semidefinite* matrix, that is,

$$\mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} \geq 0 \text{ for any } d\text{-dimensional vector } \mathbf{a}$$

To see this, observe that

$$\begin{aligned} \mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} &= \mathbf{a}^T E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]\mathbf{a} \\ &= E[\mathbf{a}^T(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T\mathbf{a}] \\ &= E[Y^2] \\ &\geq 0 \end{aligned}$$

where Y is the random variable $Y = \mathbf{a}^T(\mathbf{X} - \boldsymbol{\mu}) = \sum_{i=1}^d a_i(X_i - \mu_i)$, and we use the fact that the expectation of a squared random variable is non-negative.

Because $\boldsymbol{\Sigma}$ is also symmetric, this implies that all the eigenvalues of $\boldsymbol{\Sigma}$ are real and non-negative. In other words the d eigenvalues of $\boldsymbol{\Sigma}$ can be arranged from the largest to the smallest as follows: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$. A consequence is that the determinant of $\boldsymbol{\Sigma}$ is non-negative:

$$\det(\boldsymbol{\Sigma}) = \prod_{i=1}^d \lambda_i \geq 0 \tag{2.27}$$

Total and Generalized Variance

The total variance is given as the trace of the covariance matrix:

$$var(\mathbf{D}) = tr(\boldsymbol{\Sigma}) = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_d^2 \tag{2.28}$$

Being a sum of squares, the total variance must be non-negative.

The generalized variance is defined as the determinant of the covariance matrix, $\det(\boldsymbol{\Sigma})$, also denoted as $|\boldsymbol{\Sigma}|$. It gives a single value for the overall multivariate scatter. From Eq. (2.27) we have $\det(\boldsymbol{\Sigma}) \geq 0$.

Sample Covariance Matrix

The *sample covariance matrix* is given as

$$\widehat{\boldsymbol{\Sigma}} = E[(\mathbf{X}-\hat{\boldsymbol{\mu}})(\mathbf{X}-\hat{\boldsymbol{\mu}})^T] = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \cdots & \hat{\sigma}_{1d} \\ \hat{\sigma}_{21} & \hat{\sigma}_2^2 & \cdots & \hat{\sigma}_{2d} \\ \cdots & \cdots & \cdots & \cdots \\ \hat{\sigma}_{d1} & \hat{\sigma}_{d2} & \cdots & \hat{\sigma}_d^2 \end{pmatrix} \tag{2.29}$$

Instead of computing the sample covariance matrix element-by-element, we can obtain it via matrix operations. Let $\mathbf{Z}$ represent the centered data matrix, given as the matrix of centered attribute vectors $Z_i = X_i - \mathbf{1}\cdot\hat{\mu}_i$, where $\mathbf{1}\in\mathbb{R}^n$:

$$\mathbf{Z} = \mathbf{D} - \mathbf{1}\cdot\hat{\boldsymbol{\mu}}^T = \begin{pmatrix} | & | & & | \\ Z_1 & Z_2 & \cdots & Z_d \\ | & | & & | \end{pmatrix}$$

Alternatively, the centered data matrix can also be written in terms of the centered points $\mathbf{z}_i = \mathbf{x}_i - \hat{\boldsymbol{\mu}}$:

$$\mathbf{Z} = \mathbf{D} - \mathbf{1}\cdot\hat{\boldsymbol{\mu}}^T = \begin{pmatrix} \mathbf{x}_1^T - \hat{\boldsymbol{\mu}}^T \\ \mathbf{x}_2^T - \hat{\boldsymbol{\mu}}^T \\ \vdots \\ \mathbf{x}_n^T - \hat{\boldsymbol{\mu}}^T \end{pmatrix} = \begin{pmatrix} - & \mathbf{z}_1^T & - \\ - & \mathbf{z}_2^T & - \\ & \vdots & \\ - & \mathbf{z}_n^T & - \end{pmatrix}$$

In matrix notation, the sample covariance matrix can be written as

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n}\left(\mathbf{Z}^T\mathbf{Z}\right) = \frac{1}{n}\begin{pmatrix} Z_1^TZ_1 & Z_1^TZ_2 & \cdots & Z_1^TZ_d \\ Z_2^TZ_1 & Z_2^TZ_2 & \cdots & Z_2^TZ_d \\ \vdots & \vdots & \ddots & \vdots \\ Z_d^TZ_1 & Z_d^TZ_2 & \cdots & Z_d^TZ_d \end{pmatrix} \tag{2.30}$$

The sample covariance matrix is thus given as the pairwise *inner or dot products* of the centered attribute vectors, normalized by the sample size.

In terms of the centered points $\mathbf{z}_i$, the sample covariance matrix can also be written as a sum of rank-one matrices obtained as the *outer product* of each centered point:

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\cdot\mathbf{z}_i^T \tag{2.31}$$

Example 2.4 (Sample Mean and Covariance Matrix). Let us consider all four numeric attributes for the Iris dataset, namely `sepal length`, `sepal width`, `petal length`, and `petal width`. The multivariate sample mean vector is given as

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} 5.843 & 3.054 & 3.759 & 1.199 \end{pmatrix}^T$$

and the sample covariance matrix is given as

$$\widehat{\mathbf{\Sigma}} = \begin{pmatrix} 0.681 & -0.039 & 1.265 & 0.513 \\ -0.039 & 0.187 & -0.320 & -0.117 \\ 1.265 & -0.320 & 3.092 & 1.288 \\ 0.513 & -0.117 & 1.288 & 0.579 \end{pmatrix}$$

The sample total variance is

$$var(\mathbf{D}) = tr(\widehat{\mathbf{\Sigma}}) = 0.681 + 0.187 + 3.092 + 0.579 = 4.539$$

and the generalized variance is

$$\det(\widehat{\mathbf{\Sigma}}) = 1.853 \times 10^{-3}$$

Example 2.5 (Inner and Outer Product). To illustrate the inner and outer product–based computation of the sample covariance matrix, consider the 2-dimensional dataset

$$\mathbf{D} = \begin{pmatrix} A_1 & A_2 \\ \hline 1 & 0.8 \\ 5 & 2.4 \\ 9 & 5.5 \end{pmatrix}$$

The mean vector is as follows:

$$\hat{\mu} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} 15/3 \\ 8.7/3 \end{pmatrix} = \begin{pmatrix} 5 \\ 2.9 \end{pmatrix}$$

and the centered data matrix is then given as

$$\mathbf{Z} = \mathbf{D} - \mathbf{1} \cdot \boldsymbol{\mu}^T = \begin{pmatrix} 1 & 0.8 \\ 5 & 2.4 \\ 9 & 5.5 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 5 & 2.9 \end{pmatrix} = \begin{pmatrix} -4 & -2.1 \\ 0 & -0.5 \\ 4 & 2.6 \end{pmatrix}$$

The inner-product approach [Eq. (2.30)] to compute the sample covariance matrix gives

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n}\mathbf{Z}^T\mathbf{Z} = \frac{1}{3}\begin{pmatrix} -4 & 0 & 4 \\ -2.1 & -0.5 & 2.6 \end{pmatrix} \cdot \begin{pmatrix} -4 & -2.1 \\ 0 & -0.5 \\ 4 & 2.6 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 32 & 18.8 \\ 18.8 & 11.42 \end{pmatrix} = \begin{pmatrix} 10.67 & 6.27 \\ 6.27 & 3.81 \end{pmatrix}$$

Alternatively, the outer-product approach [Eq. (2.31)] gives

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{z}_i \cdot \mathbf{z}_i^T$$

$$= \frac{1}{3}\left[\begin{pmatrix} -4 \\ -2.1 \end{pmatrix} \cdot \begin{pmatrix} -4 & -2.1 \end{pmatrix} + \begin{pmatrix} 0 \\ -0.5 \end{pmatrix} \cdot \begin{pmatrix} 0 & -0.5 \end{pmatrix} + \begin{pmatrix} 4 \\ 2.6 \end{pmatrix} \cdot \begin{pmatrix} 4 & 2.6 \end{pmatrix}\right]$$

$$= \frac{1}{3}\left[\begin{pmatrix}16.0 & 8.4\\ 8.4 & 4.41\end{pmatrix} + \begin{pmatrix}0.0 & 0.0\\ 0.0 & 0.25\end{pmatrix} + \begin{pmatrix}16.0 & 10.4\\ 10.4 & 6.76\end{pmatrix}\right]$$

$$= \frac{1}{3}\begin{pmatrix}32.0 & 18.8\\ 18.8 & 11.42\end{pmatrix} = \begin{pmatrix}10.67 & 6.27\\ 6.27 & 3.81\end{pmatrix}$$

where the centered points $\mathbf{z}_i$ are the rows of $\mathbf{Z}$. We can see that both the inner and outer product approaches yield the same sample covariance matrix.

2.4 DATA NORMALIZATION

When analyzing two or more attributes it is often necessary to normalize the values of the attributes, especially in those cases where the values are vastly different in scale.

Range Normalization

Let X be an attribute and let $x_1, x_2, \dots, x_n$ be a random sample drawn from X. In *range normalization* each value is scaled by the sample range $\hat{r}$ of X:

$$x_i' = \frac{x_i - \min_i\{x_i\}}{\hat{r}} = \frac{x_i - \min_i\{x_i\}}{\max_i\{x_i\} - \min_i\{x_i\}}$$

After transformation the new attribute takes on values in the range $[0, 1]$.

Standard Score Normalization

In *standard score normalization*, also called z-normalization, each value is replaced by its z-score:

$$x_i' = \frac{x_i - \hat{\mu}}{\hat{\sigma}}$$

where $\hat{\mu}$ is the sample mean and $\hat{\sigma}^2$ is the sample variance of X. After transformation, the new attribute has mean $\hat{\mu}' = 0$, and standard deviation $\hat{\sigma}' = 1$.

Example 2.6. Consider the example dataset shown in Table 2.1. The attributes `Age` and `Income` have very different scales, with the latter having much larger values. Consider the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$:

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = \left\|(2, 200)^T\right\| = \sqrt{2^2 + 200^2} = \sqrt{40004} = 200.01$$

As we can observe, the contribution of `Age` is overshadowed by the value of `Income`.

The sample range for `Age` is $\hat{r} = 40 - 12 = 28$, with the minimum value 12. After range normalization, the new attribute is given as

$$\texttt{Age}' = (0, 0.071, 0.214, 0.393, 0.536, 0.571, 0.786, 0.893, 0.964, 1)^T$$

For example, for the point $\mathbf{x}_2 = (x_{21}, x_{22}) = (14, 500)$, the value $x_{21} = 14$ is transformed into

$$x_{21}' = \frac{14 - 12}{28} = \frac{2}{28} = 0.071$$

Table 2.1. Dataset for normalization

$\mathbf{x}_i$	Age (X_1)	Income (X_2)
$\mathbf{x}_1$	12	300
$\mathbf{x}_2$	14	500
$\mathbf{x}_3$	18	1000
$\mathbf{x}_4$	23	2000
$\mathbf{x}_5$	27	3500
$\mathbf{x}_6$	28	4000
$\mathbf{x}_7$	34	4300
$\mathbf{x}_8$	37	6000
$\mathbf{x}_9$	39	2500
$\mathbf{x}_{10}$	40	2700

Likewise, the sample range for `Income` is $6000 - 300 = 5700$, with a minimum value of 300; `Income` is therefore transformed into

$$\texttt{Income}' = (0, 0.035, 0.123, 0.298, 0.561, 0.649, 0.702, 1, 0.386, 0.421)^T$$

so that $x_{22} = 0.035$. The distance between $\mathbf{x}_1$ and $\mathbf{x}_2$ after range normalization is given as

$$\left\|\mathbf{x}_1' - \mathbf{x}_2'\right\| = \left\|(0,0)^T - (0.071, 0.035)^T\right\| = \left\|(-0.071, -0.035)^T\right\| = 0.079$$

We can observe that `Income` no longer skews the distance.

For z-normalization, we first compute the mean and standard deviation of both attributes:

	Age	Income
$\hat{\mu}$	27.2	2680
$\hat{\sigma}$	9.77	1726.15

`Age` is transformed into

$$\texttt{Age}' = (-1.56, -1.35, -0.94, -0.43, -0.02, 0.08, 0.70, 1.0, 1.21, 1.31)^T$$

For instance, the value $x_{21} = 14$, for the point $\mathbf{x}_2 = (x_{21}, x_{22}) = (14, 500)$, is transformed as

$$x_{21}' = \frac{14 - 27.2}{9.77} = -1.35$$

Likewise, `Income` is transformed into

$$\texttt{Income}' = (-1.38, -1.26, -0.97, -0.39, 0.48, 0.77, 0.94, 1.92, -0.10, 0.01)^T$$

so that $x_{22} = -1.26$. The distance between $\mathbf{x}_1$ and $\mathbf{x}_2$ after z-normalization is given as

$$\left\|\mathbf{x}_1' - \mathbf{x}_2'\right\| = \left\|(-1.56, -1.38)^T - (1.35, -1.26)^T\right\| = \left\|(-0.18, -0.12)^T\right\| = 0.216$$

2.5 NORMAL DISTRIBUTION

The normal distribution is one of the most important probability density functions, especially because many physically observed variables follow an approximately normal distribution. Furthermore, the sampling distribution of the mean of any arbitrary probability distribution follows a normal distribution. The normal distribution also plays an important role as the parametric distribution of choice in clustering, density estimation, and classification.

2.5.1 Univariate Normal Distribution

A random variable X has a normal distribution, with the parameters mean μ and variance σ^2, if the probability density function of X is given as follows:

$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

The term $(x-\mu)^2$ measures the distance of a value x from the mean μ of the distribution, and thus the probability density decreases exponentially as a function of the distance from the mean. The maximum value of the density occurs at the mean value $x = \mu$, given as $f(\mu) = \frac{1}{\sqrt{2\pi\sigma^2}}$, which is inversely proportional to the standard deviation σ of the distribution.

Example 2.7. Figure 2.5 plots the standard normal distribution, which has the parameters $\mu = 0$ and $\sigma^2 = 1$. The normal distribution has a characteristic *bell* shape, and it is symmetric about the mean. The figure also shows the effect of different values of standard deviation on the shape of the distribution. A smaller value (e.g., $\sigma = 0.5$) results in a more "peaked" distribution that decays faster, whereas a larger value (e.g., $\sigma = 2$) results in a flatter distribution that decays slower. Because the normal distribution is symmetric, the mean μ is also the median, as well as the mode, of the distribution.

Probability Mass

Given an interval $[a,b]$ the probability mass of the normal distribution within that interval is given as

$$P(a \le x \le b) = \int_a^b f(x|\,\mu,\sigma^2)\,dx$$

In particular, we are often interested in the probability mass concentrated within k standard deviations from the mean, that is, for the interval $[\mu-k\sigma, \mu+k\sigma]$, which can be computed as

$$P\big(\mu-k\sigma \le x \le \mu+k\sigma\big) = \frac{1}{\sqrt{2\pi}\sigma}\int_{\mu-k\sigma}^{\mu+k\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx$$

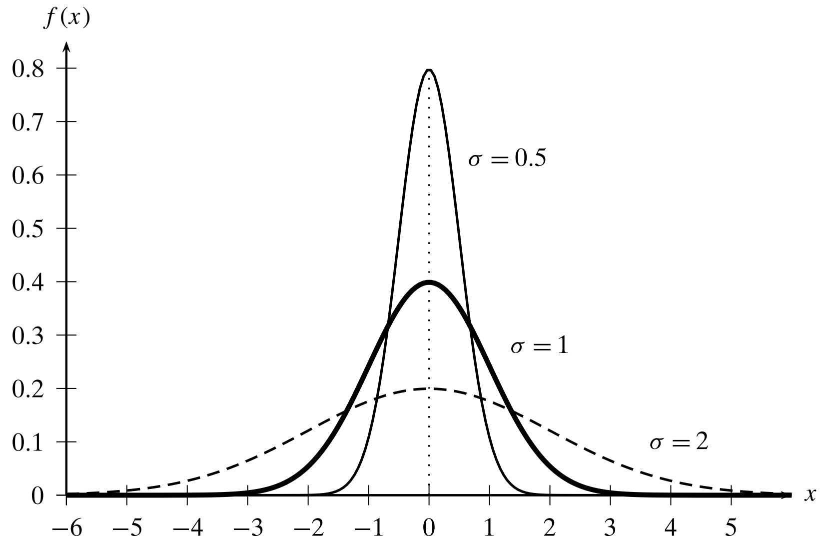

Figure 2.5. Normal distribution: $\mu = 0$, and different variances.

Via a change of variable $z = \frac{x-\mu}{\sigma}$, we get an equivalent formulation in terms of the standard normal distribution:

$$\begin{aligned} P(-k \le z \le k) &= \frac{1}{\sqrt{2\pi}} \int_{-k}^{k} e^{-\frac{1}{2}z^2} dz \\ &= \frac{2}{\sqrt{2\pi}} \int_{0}^{k} e^{-\frac{1}{2}z^2} dz \end{aligned}$$

The last step follows from the fact that $e^{-\frac{1}{2}z^2}$ is symmetric, and thus the integral over the range $[-k, k]$ is equivalent to 2 times the integral over the range $[0, k]$. Finally, via another change of variable $t = \frac{z}{\sqrt{2}}$, we get

$$P(-k \le z \le k) = 2 \cdot P\big(0 \le t \le k/\sqrt{2}\big) = \frac{2}{\sqrt{\pi}} \int_{0}^{k/\sqrt{2}} e^{-t^2} dt = \mathrm{erf}\Big(k/\sqrt{2}\Big) \tag{2.32}$$

where erf is the *Gauss error function*, defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{0}^{x} e^{-t^2} dt$$

Using Eq. (2.32) we can compute the probability mass within k standard deviations of the mean. In particular, for $k = 1$, we have

$$P(\mu - \sigma \le x \le \mu + \sigma) = \mathrm{erf}(1/\sqrt{2}) = 0.6827$$

which means that 68.27% of all points lie within 1 standard deviation from the mean. For $k = 2$, we have $\text{erf}(2/\sqrt{2}) = 0.9545$, and for $k = 3$ we have $\text{erf}(3/\sqrt{2}) = 0.9973$. Thus, almost the entire probability mass (i.e., 99.73%) of a normal distribution is within $\pm 3\sigma$ from the mean μ.

2.5.2 Multivariate Normal Distribution

Given the d-dimensional vector random variable $\mathbf{X} = (X_1, X_2, \ldots, X_d)^T$, we say that $\mathbf{X}$ has a multivariate normal distribution, with the parameters mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, if its joint multivariate probability density function is given as follows:

$$f(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(\sqrt{2\pi})^d \sqrt{|\boldsymbol{\Sigma}|}} \exp\left\{ -\frac{(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}{2} \right\} \tag{2.33}$$

where $|\boldsymbol{\Sigma}|$ is the determinant of the covariance matrix. As in the univariate case, the term

$$(\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \tag{2.34}$$

measures the distance, called the *Mahalanobis distance*, of the point $\mathbf{x}$ from the mean $\boldsymbol{\mu}$ of the distribution, taking into account all of the variance–covariance information between the attributes. The Mahalanobis distance is a generalization of Euclidean distance because if we set $\boldsymbol{\Sigma} = \mathbf{I}$, where $\mathbf{I}$ is the $d \times d$ identity matrix (with diagonal elements as 1's and off-diagonal elements as 0's), we get

$$(\mathbf{x}_i - \boldsymbol{\mu})^T \mathbf{I}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) = \|\mathbf{x}_i - \boldsymbol{\mu}\|^2$$

The Euclidean distance thus ignores the covariance information between the attributes, whereas the Mahalanobis distance explicitly takes it into consideration.

The *standard multivariate normal distribution* has parameters $\mu = \mathbf{0}$ and $\boldsymbol{\Sigma} = \mathbf{I}$. Figure 2.6a plots the probability density of the standard bivariate ($d = 2$) normal distribution, with parameters

$$\boldsymbol{\mu} = \mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and

$$\boldsymbol{\Sigma} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This corresponds to the case where the two attributes are independent, and both follow the standard normal distribution. The symmetric nature of the standard normal distribution can be clearly seen in the contour plot shown in Figure 2.6b. Each level curve represents the set of points $\mathbf{x}$ with a fixed density value $f(\mathbf{x})$.

Geometry of the Multivariate Normal

Let us consider the geometry of the multivariate normal distribution for an arbitrary mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Compared to the standard normal distribution, we can expect the density contours to be shifted, scaled, and rotated. The shift or translation comes from the fact that the mean $\boldsymbol{\mu}$ is not necessarily the origin $\mathbf{0}$. The

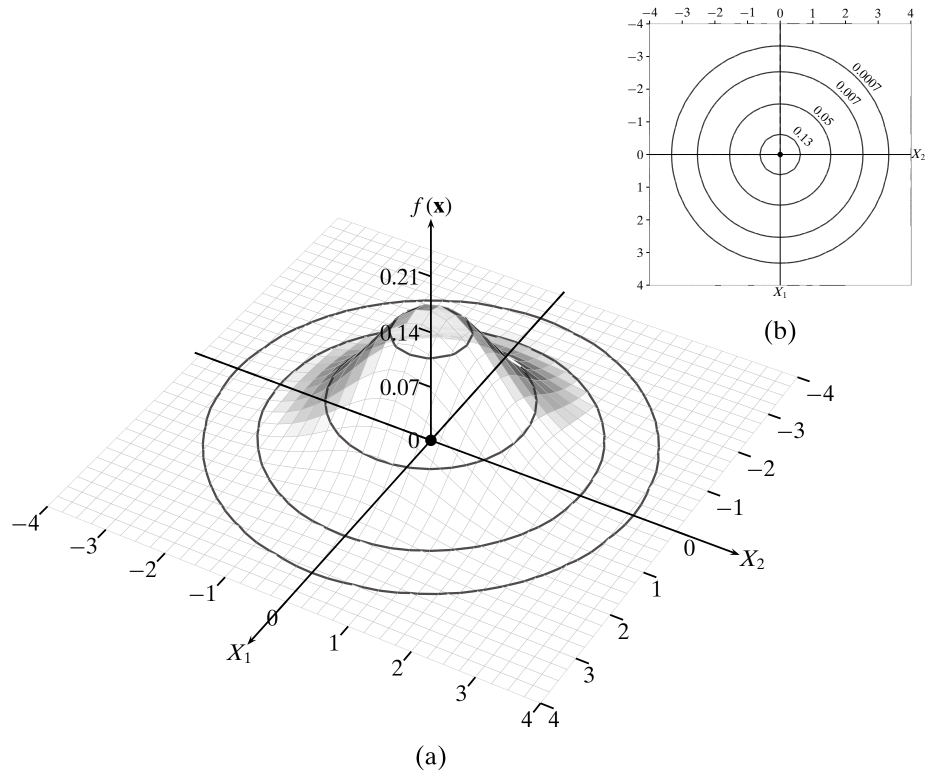

Figure 2.6. (a) Standard bivariate normal density and (b) its contour plot. Parameters: $\boldsymbol{\mu} = (0,0)^T$, $\boldsymbol{\Sigma} = \mathbf{I}$.

scaling or skewing is a result of the attribute variances, and the rotation is a result of the covariances.

The shape or geometry of the normal distribution becomes clear by considering the eigen-decomposition of the covariance matrix. Recall that $\boldsymbol{\Sigma}$ is a $d \times d$ symmetric positive semidefinite matrix. The eigenvector equation for $\boldsymbol{\Sigma}$ is given as

$$\boldsymbol{\Sigma}\mathbf{u}_i = \lambda_i \mathbf{u}_i$$

Here λ_i is an eigenvalue of $\boldsymbol{\Sigma}$ and the vector $\mathbf{u}_i \in \mathbb{R}^d$ is the eigenvector corresponding to λ_i. Because $\boldsymbol{\Sigma}$ is symmetric and positive semidefinite it has d real and non-negative eigenvalues, which can be arranged in order from the largest to the smallest as follows: $\lambda_1 \ge \lambda_2 \ge \ldots \lambda_d \ge 0$. The diagonal matrix $\boldsymbol{\Lambda}$ is used to record these eigenvalues:

$$\boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_d \end{pmatrix}$$

Further, the eigenvectors are unit vectors (normal) and are mutually orthogonal, that is, they are orthonormal:

$$\mathbf{u}_i^T\mathbf{u}_i = 1 \quad \text{for all } i$$
$$\mathbf{u}_i^T\mathbf{u}_j = 0 \quad \text{for all } i \neq j$$

The eigenvectors can be put together into an orthogonal matrix $\mathbf{U}$, defined as a matrix with normal and mutually orthogonal columns:

$$\mathbf{U} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_d \\ | & | & & | \end{pmatrix}$$

The eigen-decomposition of $\mathbf{\Sigma}$ can then be expressed compactly as follows:

$$\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$$

This equation can be interpreted geometrically as a change in basis vectors. From the original d dimensions corresponding to the d attributes X_j, we derive d new dimensions $\mathbf{u}_i$. $\mathbf{\Sigma}$ is the covariance matrix in the original space, whereas $\mathbf{\Lambda}$ is the covariance matrix in the new coordinate space. Because $\mathbf{\Lambda}$ is a diagonal matrix, we can immediately conclude that after the transformation, each new dimension $\mathbf{u}_i$ has variance λ_i, and further that all covariances are zero. In other words, in the new space, the normal distribution is axis aligned (has no rotation component), but is skewed in each axis proportional to the eigenvalue λ_i, which represents the variance along that dimension (further details are given in Section 7.2.4).

Total and Generalized Variance

The determinant of the covariance matrix is is given as $\det(\mathbf{\Sigma}) = \prod_{i=1}^d \lambda_i$. Thus, the generalized variance of $\mathbf{\Sigma}$ is the product of its eigenvalues.

Given the fact that the trace of a square matrix is invariant to similarity transformation, such as a change of basis, we conclude that the total variance $var(\mathbf{D})$ for a dataset $\mathbf{D}$ is invariant, that is,

$$var(\mathbf{D}) = tr(\mathbf{\Sigma}) = \sum_{i=1}^d \sigma_i^2 = \sum_{i=1}^d \lambda_i = tr(\mathbf{\Lambda})$$

In other words $\sigma_1^2 + \cdots + \sigma_d^2 = \lambda_1 + \cdots + \lambda_d$.

Example 2.8 (Bivariate Normal Density). Treating attributes `sepal length` (X_1) and `sepal width` (X_2) in the Iris dataset (see Table 1.1) as continuous random variables, we can define a continuous bivariate random variable $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$. Assuming that $\mathbf{X}$ follows a bivariate normal distribution, we can estimate its parameters from the sample. The sample mean is given as

$$\hat{\boldsymbol{\mu}} = (5.843, 3.054)^T$$

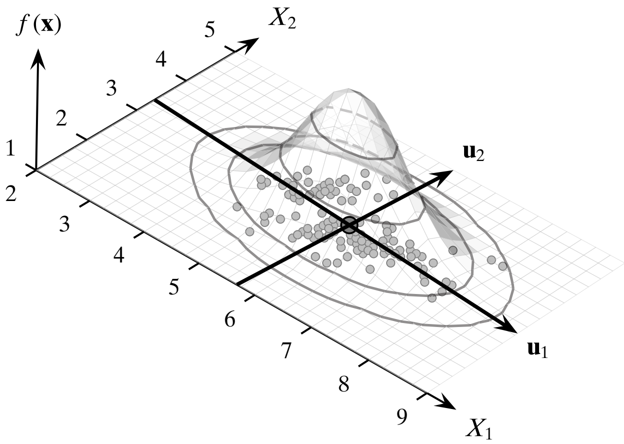

Figure 2.7. Iris: `sepal length` and `sepal width`, bivariate normal density and contours.

and the sample covariance matrix is given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} 0.681 & -0.039 \\ -0.039 & 0.187 \end{pmatrix}$$

The plot of the bivariate normal density for the two attributes is shown in Figure 2.7. The figure also shows the contour lines and the data points.

Consider the point $\mathbf{x}_2 = (6.9, 3.1)^T$. We have

$$\mathbf{x}_2 - \hat{\boldsymbol{\mu}} = \begin{pmatrix} 6.9 \\ 3.1 \end{pmatrix} - \begin{pmatrix} 5.843 \\ 3.054 \end{pmatrix} = \begin{pmatrix} 1.057 \\ 0.046 \end{pmatrix}$$

The Mahalanobis distance between $\mathbf{x}_2$ and $\hat{\boldsymbol{\mu}}$ is

$$\begin{aligned}
(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^T \, \widehat{\boldsymbol{\Sigma}}^{-1} \, (\mathbf{x}_i - \hat{\boldsymbol{\mu}}) &= \begin{pmatrix} 1.057 & 0.046 \end{pmatrix} \begin{pmatrix} 0.681 & -0.039 \\ -0.039 & 0.187 \end{pmatrix}^{-1} \begin{pmatrix} 1.057 \\ 0.046 \end{pmatrix} \\
&= \begin{pmatrix} 1.057 & 0.046 \end{pmatrix} \begin{pmatrix} 1.486 & 0.31 \\ 0.31 & 5.42 \end{pmatrix} \begin{pmatrix} 1.057 \\ 0.046 \end{pmatrix} \\
&= 1.701
\end{aligned}$$

whereas the squared Euclidean distance between them is

$$\|(\mathbf{x}_2 - \hat{\boldsymbol{\mu}})\|^2 = \begin{pmatrix} 1.057 & 0.046 \end{pmatrix} \begin{pmatrix} 1.057 \\ 0.046 \end{pmatrix} = 1.119$$

The eigenvalues and the corresponding eigenvectors of $\widehat{\boldsymbol{\Sigma}}$ are as follows:

$$\lambda_1 = 0.684 \qquad \mathbf{u}_1 = (-0.997, 0.078)^T$$

$$\lambda_2 = 0.184 \qquad \mathbf{u}_2 = (-0.078, -0.997)^T$$

These two eigenvectors define the new axes in which the covariance matrix is given as

$$\mathbf{\Lambda} = \begin{pmatrix} 0.684 & 0 \\ 0 & 0.184 \end{pmatrix}$$

The angle between the original axes $\mathbf{e}_1 = (1,0)^T$ and $\mathbf{u}_1$ specifies the rotation angle for the multivariate normal:

$$\cos\theta = \mathbf{e}_1^T \mathbf{u}_1 = -0.997$$
$$\theta = \cos^{-1}(-0.997) = 175.5^\circ$$

Figure 2.7 illustrates the new coordinate axes and the new variances. We can see that in the original axes, the contours are only slightly rotated by angle 175.5° (or -4.5°).

2.6 FURTHER READING

There are several good textbooks that cover the topics discussed in this chapter in more depth; see Evans and Rosenthal (2011) and Wasserman (2004) and Rencher and Christensen (2012).

Evans, M. and Rosenthal, J. (2011). *Probability and Statistics: The Science of Uncertainty*. 2nd ed. New York: W. H. Freeman.

Rencher, A. C. and Christensen, W. F. (2012). *Methods of multivariate analysis*. 3rd ed. Hoboken, NJ: John Wiley & Sons.

Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. New York: Springer Science + Business Media.

2.7 EXERCISES

Q1. True or False:

(a) Mean is robust against outliers.

(b) Median is robust against outliers.

(c) Standard deviation is robust against outliers.

Q2. Let X and Y be two random variables, denoting age and weight, respectively. Consider a random sample of size $n = 20$ from these two variables

$$X = (69, 74, 68, 70, 72, 67, 66, 70, 76, 68, 72, 79, 74, 67, 66, 71, 74, 75, 75, 76)$$
$$Y = (153, 175, 155, 135, 172, 150, 115, 137, 200, 130, 140, 265, 185, 112, 140, 150, 165, 185, 210, 220)$$

(a) Find the mean, median, and mode for X.

(b) What is the variance for Y?

(c) Plot the normal distribution for X.
(d) What is the probability of observing an age of 80 or higher?
(e) Find the 2-dimensional mean $\hat{\boldsymbol{\mu}}$ and the covariance matrix $\widehat{\boldsymbol{\Sigma}}$ for these two variables.
(f) What is the correlation between age and weight?
(g) Draw a scatterplot to show the relationship between age and weight.

Q3. Show that the identity in Eq. (2.15) holds, that is,

$$\sum_{i=1}^{n}(x_i-\mu)^2 = n(\hat{\mu}-\mu)^2 + \sum_{i=1}^{n}(x_i-\hat{\mu})^2$$

Q4. Prove that if x_i are independent random variables, then

$$var\left(\sum_{i=1}^{n} x_i\right) = \sum_{i=1}^{n} var(x_i)$$

This fact was used in Eq. (2.12).

Q5. Define a measure of deviation called *mean absolute deviation* for a random variable X as follows:

$$\frac{1}{n}\sum_{i=1}^{n}|x_i-\mu|$$

Is this measure robust? Why or why not?

Q6. Prove that the expected value of a vector random variable $\mathbf{X}=(X_1,X_2)^T$ is simply the vector of the expected value of the individual random variables X_1 and X_2 as given in Eq. (2.18).

Q7. Show that the correlation [Eq. (2.23)] between any two random variables X_1 and X_2 lies in the range $[-1,1]$.

Q8. Given the dataset in Table 2.2, compute the covariance matrix and the generalized variance.

Table 2.2. Dataset for Q8

	X_1	X_2	X_3
$\mathbf{x}_1$	17	17	12
$\mathbf{x}_2$	11	9	13
$\mathbf{x}_3$	11	8	19

Q9. Show that the outer-product in Eq. (2.31) for the sample covariance matrix is equivalent to Eq. (2.29).

Q10. Assume that we are given two univariate normal distributions, N_A and N_B, and let their mean and standard deviation be as follows: $\mu_A=4, \sigma_A=1$ and $\mu_B=8, \sigma_B=2$.
(a) For each of the following values $x_i \in \{5,6,7\}$ find out which is the more likely normal distribution to have produced it.
(b) Derive an expression for the point for which the probability of having been produced by both the normals is the same.

Q11. Consider Table 2.3. Assume that both the attributes X and Y are numeric, and the table represents the entire population. If we know that the correlation between X and Y is zero, what can you infer about the values of Y?

Table 2.3. Dataset for Q11

X	Y
1	a
0	b
1	c
0	a
0	c

Q12. Under what conditions will the covariance matrix $\boldsymbol{\Sigma}$ be identical to the correlation matrix, whose (i, j) entry gives the correlation between attributes X_i and X_j? What can you conclude about the two variables?

CHAPTER 3 Categorical Attributes

In this chapter we present methods to analyze categorical attributes. Because categorical attributes have only symbolic values, many of the arithmetic operations cannot be performed directly on the symbolic values. However, we can compute the frequencies of these values and use them to analyze the attributes.

3.1 UNIVARIATE ANALYSIS

We assume that the data consists of values for a single categorical attribute, X. Let the domain of X consist of m symbolic values $dom(X) = \{a_1, a_2, \dots, a_m\}$. The data $\mathbf{D}$ is thus an $n \times 1$ symbolic data matrix given as

$$\mathbf{D} = \begin{pmatrix} X \\ \hline x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

where each point $x_i \in dom(X)$.

3.1.1 Bernoulli Variable

Let us first consider the case when the categorical attribute X has domain $\{a_1, a_2\}$, with $m = 2$. We can model X as a Bernoulli random variable, which takes on two distinct values, 1 and 0, according to the mapping

$$X(v) = \begin{cases} 1 & \text{if } v = a_1 \\ 0 & \text{if } v = a_2 \end{cases}$$

The probability mass function (PMF) of X is given as

$$P(X = x) = f(x) = \begin{cases} p_1 & \text{if } x = 1 \\ p_0 & \text{if } x = 0 \end{cases}$$

where p_1 and p_0 are the parameters of the distribution, which must satisfy the condition

$$p_1 + p_0 = 1$$

Because there is only one free parameter, it is customary to denote $p_1 = p$, from which it follows that $p_0 = 1 - p$. The PMF of Bernoulli random variable X can then be written compactly as

$$P(X = x) = f(x) = p^x(1-p)^{1-x}$$

We can see that $P(X = 1) = p^1(1-p)^0 = p$ and $P(X = 0) = p^0(1-p)^1 = 1-p$, as desired.

Mean and Variance

The expected value of X is given as

$$\mu = E[X] = 1 \cdot p + 0 \cdot (1-p) = p$$

and the variance of X is given as

$$\begin{aligned}\sigma^2 = var(X) &= E[X^2] - (E[X])^2 \\ &= (1^2 \cdot p + 0^2 \cdot (1-p)) - p^2 = p - p^2 = p(1-p) \qquad (3.1)\end{aligned}$$

Sample Mean and Variance

To estimate the parameters of the Bernoulli variable X, we assume that each symbolic point has been mapped to its binary value. Thus, the set $\{x_1, x_2, \ldots, x_n\}$ is assumed to be a random sample drawn from X (i.e., each x_i is IID with X).

The sample mean is given as

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{n_1}{n} = \hat{p} \qquad (3.2)$$

where n_1 is the number of points with $x_i = 1$ in the random sample (equal to the number of occurrences of symbol a_1).

Let $n_0 = n - n_1$ denote the number of points with $x_i = 0$ in the random sample. The sample variance is given as

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2 \\ &= \frac{n_1}{n}(1-\hat{p})^2 + \frac{n-n_1}{n}(-\hat{p})^2 \\ &= \hat{p}(1-\hat{p})^2 + (1-\hat{p})\hat{p}^2 \\ &= \hat{p}(1-\hat{p})(1-\hat{p}+\hat{p}) \\ &= \hat{p}(1-\hat{p})\end{aligned}$$

The sample variance could also have been obtained directly from Eq. (3.1), by substituting $\hat{p}$ for p.

Example 3.1. Consider the `sepal length` attribute (X_1) for the Iris dataset in Table 1.1. Let us define an Iris flower as `Long` if its sepal length is in the range $[7, \infty]$, and `Short` if its sepal length is in the range $[-\infty, 7)$. Then X_1 can be treated as a categorical attribute with domain $\{\texttt{Long}, \texttt{Short}\}$. From the observed sample of size $n = 150$, we find 13 long Irises. The sample mean of X_1 is

$$\hat{\mu} = \hat{p} = 13/150 = 0.087$$

and its variance is

$$\hat{\sigma}^2 = \hat{p}(1 - \hat{p}) = 0.087(1 - 0.087) = 0.087 \cdot 0.913 = 0.079$$

Binomial Distribution: Number of Occurrences

Given the Bernoulli variable X, let $\{x_1, x_2, \ldots, x_n\}$ denote a random sample of size n drawn from X. Let N be the random variable denoting the number of occurrences of the symbol a_1 (value $X = 1$) in the sample. N has a binomial distribution, given as

$$f(N = n_1 \mid n, p) = \binom{n}{n_1} p^{n_1} (1 - p)^{n - n_1} \tag{3.3}$$

In fact, N is the sum of the n independent Bernoulli random variables x_i IID with X, that is, $N = \sum_{i=1}^{n} x_i$. By linearity of expectation, the mean or expected number of occurrences of symbol a_1 is given as

$$\mu_N = E[N] = E\left[\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} E[x_i] = \sum_{i=1}^{n} p = np$$

Because x_i are all independent, the variance of N is given as

$$\sigma_N^2 = var(N) = \sum_{i=1}^{n} var(x_i) = \sum_{i=1}^{n} p(1 - p) = np(1 - p)$$

Example 3.2. Continuing with Example 3.1, we can use the estimated parameter $\hat{p} = 0.087$ to compute the expected number of occurrences N of `Long` sepal length Irises via the binomial distribution:

$$E[N] = n\hat{p} = 150 \cdot 0.087 = 13$$

In this case, because p is estimated from the sample via $\hat{p}$, it is not surprising that the expected number of occurrences of long Irises coincides with the actual occurrences. However, what is more interesting is that we can compute the variance in the number of occurrences:

$$var(N) = n\hat{p}(1 - \hat{p}) = 150 \cdot 0.079 = 11.9$$

As the sample size increases, the binomial distribution given in Eq. 3.3 tends to a normal distribution with $\mu = 13$ and $\sigma = \sqrt{11.9} = 3.45$ for our example. Thus, with confidence greater than 95% we can claim that the number of occurrences of a_1 will lie in the range $\mu \pm 2\sigma = [9.55, 16.45]$, which follows from the fact that for a normal distribution 95.45% of the probability mass lies within two standard deviations from the mean (see Section 2.5.1).

3.1.2 Multivariate Bernoulli Variable

We now consider the general case when X is a categorical attribute with domain $\{a_1, a_2, \dots, a_m\}$. We can model X as an m-dimensional Bernoulli random variable $\mathbf{X} = (A_1, A_2, \dots, A_m)^T$, where each A_i is a Bernoulli variable with parameter p_i denoting the probability of observing symbol a_i. However, because X can assume only one of the symbolic values at any one time, if $X = a_i$, then $A_i = 1$, and $A_j = 0$ for all $j \neq i$. The range of the random variable $\mathbf{X}$ is thus the set $\{0,1\}^m$, with the further restriction that if $X = a_i$, then $\mathbf{X} = \mathbf{e}_i$, where $\mathbf{e}_i$ is the ith standard basis vector $\mathbf{e}_i \in \mathbb{R}^m$ given as

$$\mathbf{e}_i = (\overbrace{0, \dots, 0}^{i-1}, 1, \overbrace{0, \dots, 0}^{m-i})^T$$

In $\mathbf{e}_i$, only the ith element is 1 ($e_{ii} = 1$), whereas all other elements are zero ($e_{ij} = 0, \forall j \neq i$).

This is precisely the definition of a *multivariate Bernoulli variable*, which is a generalization of a Bernoulli variable from two outcomes to m outcomes. We thus model the categorical attribute X as a multivariate Bernoulli variable $\mathbf{X}$ defined as

$$\mathbf{X}(v) = \mathbf{e}_i \text{ if } v = a_i$$

The range of $\mathbf{X}$ consists of m distinct vector values $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$, with the PMF of $\mathbf{X}$ given as

$$P(\mathbf{X} = \mathbf{e}_i) = f(\mathbf{e}_i) = p_i$$

where p_i is the probability of observing value a_i. These parameters must satisfy the condition

$$\sum_{i=1}^{m} p_i = 1$$

The PMF can be written compactly as follows:

$$P(\mathbf{X} = \mathbf{e}_i) = f(\mathbf{e}_i) = \prod_{j=1}^{m} p_j^{e_{ij}} \tag{3.4}$$

Because $e_{ii} = 1$, and $e_{ij} = 0$ for $j \neq i$, we can see that, as expected, we have

$$f(\mathbf{e}_i) = \prod_{j=1}^{m} p_j^{e_{ij}} = p_1^{e_{i0}} \times \cdots p_i^{e_{ii}} \cdots \times p_m^{e_{im}} = p_1^0 \times \cdots p_i^1 \cdots \times p_m^0 = p_i$$

Table 3.1. Discretized `sepal length` attribute

Bins	Domain	Counts
$[4.3, 5.2]$	Very Short (a_1)	$n_1 = 45$
$(5.2, 6.1]$	Short (a_2)	$n_2 = 50$
$(6.1, 7.0]$	Long (a_3)	$n_3 = 43$
$(7.0, 7.9]$	Very Long (a_4)	$n_4 = 12$

Example 3.3. Let us consider the `sepal length` attribute (X_1) for the Iris dataset shown in Table 1.2. We divide the sepal length into four equal-width intervals, and give each interval a name as shown in Table 3.1. We consider X_1 as a categorical attribute with domain

$$\{a_1 = \texttt{VeryShort}, a_2 = \texttt{Short}, a_3 = \texttt{Long}, a_4 = \texttt{VeryLong}\}$$

We model the categorical attribute X_1 as a multivariate Bernoulli variable $\mathbf{X}$, defined as

$$\mathbf{X}(v) = \begin{cases} \mathbf{e}_1 = (1,0,0,0) & \text{if } v = a_1 \\ \mathbf{e}_2 = (0,1,0,0) & \text{if } v = a_2 \\ \mathbf{e}_3 = (0,0,1,0) & \text{if } v = a_3 \\ \mathbf{e}_4 = (0,0,0,1) & \text{if } v = a_4 \end{cases}$$

For example, the symbolic point $x_1 = \texttt{Short} = a_2$ is represented as the vector $(0,1,0,0)^T = \mathbf{e}_2$.

Mean

The mean or expected value of $\mathbf{X}$ can be obtained as

$$\boldsymbol{\mu} = E[\mathbf{X}] = \sum_{i=1}^{m} \mathbf{e}_i f(\mathbf{e}_i) = \sum_{i=1}^{m} \mathbf{e}_i p_i = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} p_1 + \cdots + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} p_m = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_m \end{pmatrix} = \mathbf{p} \tag{3.5}$$

Sample Mean

Assume that each symbolic point $x_i \in \mathbf{D}$ is mapped to the variable $\mathbf{x}_i = \mathbf{X}(x_i)$. The mapped dataset $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ is then assumed to be a random sample IID with $\mathbf{X}$. We can compute the sample mean by placing a probability mass of $\frac{1}{n}$ at each point

$$\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i = \sum_{i=1}^{m} \frac{n_i}{n} \mathbf{e}_i = \begin{pmatrix} n_1/n \\ n_2/n \\ \vdots \\ n_m/n \end{pmatrix} = \begin{pmatrix} \hat{p}_1 \\ \hat{p}_2 \\ \vdots \\ \hat{p}_m \end{pmatrix} = \hat{\mathbf{p}} \tag{3.6}$$

where n_i is the number of occurrences of the vector value $\mathbf{e}_i$ in the sample, which is equivalent to the number of occurrences of the symbol a_i. Furthermore, we have

Figure 3.1. Probability mass function: `sepal length`.

$\sum_{i=1}^{m} n_i = n$, which follows from the fact that $\mathbf{X}$ can take on only m distinct values $\mathbf{e}_i$, and the counts for each value must add up to the sample size n.

Example 3.4 (Sample Mean). Consider the observed counts n_i for each of the values a_i ($\mathbf{e}_i$) of the discretized `sepal length` attribute, shown in Table 3.1. Because the total sample size is $n = 150$, from these we can obtain the estimates $\hat{p}_i$ as follows:

$$\hat{p}_1 = 45/150 = 0.3$$
$$\hat{p}_2 = 50/150 = 0.333$$
$$\hat{p}_3 = 43/150 = 0.287$$
$$\hat{p}_4 = 12/150 = 0.08$$

The PMF for $\mathbf{X}$ is plotted in Figure 3.1, and the sample mean for $\mathbf{X}$ is given as

$$\hat{\boldsymbol{\mu}} = \hat{\mathbf{p}} = \begin{pmatrix} 0.3 \\ 0.333 \\ 0.287 \\ 0.08 \end{pmatrix}$$

Covariance Matrix

Recall that an m-dimensional multivariate Bernoulli variable is simply a vector of m Bernoulli variables. For instance, $\mathbf{X} = (A_1, A_2, \dots, A_m)^T$, where A_i is the Bernoulli variable corresponding to symbol a_i. The variance–covariance information between the constituent Bernoulli variables yields a covariance matrix for $\mathbf{X}$.

Let us first consider the variance along each Bernoulli variable A_i. By Eq. (3.1), we immediately have

$$\sigma_i^2 = var(A_i) = p_i(1 - p_i)$$

Next consider the covariance between A_i and A_j. Utilizing the identity in Eq. (2.21), we have

$$\sigma_{ij} = E[A_i A_j] - E[A_i] \cdot E[A_j] = 0 - p_i p_j = -p_i p_j$$

which follows from the fact that $E[A_i A_j] = 0$, as A_i and A_j cannot both be 1 at the same time, and thus their product $A_i A_j = 0$. This same fact leads to the negative relationship between A_i and A_j. What is interesting is that the degree of negative association is proportional to the product of the mean values for A_i and A_j.

From the preceding expressions for variance and covariance, the $m \times m$ covariance matrix for $\mathbf{X}$ is given as

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{12} & \sigma_2^2 & \dots & \sigma_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1m} & \sigma_{2m} & \dots & \sigma_m^2 \end{pmatrix} = \begin{pmatrix} p_1(1-p_1) & -p_1 p_2 & \cdots & -p_1 p_m \\ -p_1 p_2 & p_2(1-p_2) & \cdots & -p_2 p_m \\ \vdots & \vdots & \ddots & \vdots \\ -p_1 p_m & -p_2 p_m & \cdots & p_m(1-p_m) \end{pmatrix}$$

Notice how each row in $\mathbf{\Sigma}$ sums to zero. For example, for row i, we have

$$-p_i p_1 - p_i p_2 - \cdots + p_i(1 - p_i) - \cdots - p_i p_m = p_i - p_i \sum_{j=1}^{m} p_j = p_i - p_i = 0 \tag{3.7}$$

Because $\mathbf{\Sigma}$ is symmetric, it follows that each column also sums to zero.

Define $\mathbf{P}$ as the $m \times m$ diagonal matrix:

$$\mathbf{P} = \text{diag}(\mathbf{p}) = \text{diag}(p_1, p_2, \dots, p_m) = \begin{pmatrix} p_1 & 0 & \cdots & 0 \\ 0 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & p_m \end{pmatrix}$$

We can compactly write the covariance matrix of $\mathbf{X}$ as

$$\mathbf{\Sigma} = \mathbf{P} - \mathbf{p} \cdot \mathbf{p}^T \tag{3.8}$$

Sample Covariance Matrix

The sample covariance matrix can be obtained from Eq. (3.8) in a straightforward manner:

$$\widehat{\mathbf{\Sigma}} = \widehat{\mathbf{P}} - \hat{\mathbf{p}} \cdot \hat{\mathbf{p}}^T \tag{3.9}$$

where $\widehat{\mathbf{P}} = \text{diag}(\hat{\mathbf{p}})$, and $\hat{\mathbf{p}} = \hat{\boldsymbol{\mu}} = (\hat{p}_1, \hat{p}_2, \dots, \hat{p}_m)^T$ denotes the empirical probability mass function for $\mathbf{X}$.

Example 3.5. Returning to the discretized `sepal length` attribute in Example 3.4, we have $\hat{\boldsymbol{\mu}} = \hat{\mathbf{p}} = (0.3, 0.333, 0.287, 0.08)^T$. The sample covariance matrix is given as

$$\begin{aligned}
\widehat{\boldsymbol{\Sigma}} &= \widehat{\mathbf{P}} - \hat{\mathbf{p}} \cdot \hat{\mathbf{p}}^T \\
&= \begin{pmatrix} 0.3 & 0 & 0 & 0 \\ 0 & 0.333 & 0 & 0 \\ 0 & 0 & 0.287 & 0 \\ 0 & 0 & 0 & 0.08 \end{pmatrix} - \begin{pmatrix} 0.3 \\ 0.333 \\ 0.287 \\ 0.08 \end{pmatrix} \begin{pmatrix} 0.3 & 0.333 & 0.287 & 0.08 \end{pmatrix} \\
&= \begin{pmatrix} 0.3 & 0 & 0 & 0 \\ 0 & 0.333 & 0 & 0 \\ 0 & 0 & 0.287 & 0 \\ 0 & 0 & 0 & 0.08 \end{pmatrix} - \begin{pmatrix} 0.09 & 0.1 & 0.086 & 0.024 \\ 0.1 & 0.111 & 0.096 & 0.027 \\ 0.086 & 0.096 & 0.082 & 0.023 \\ 0.024 & 0.027 & 0.023 & 0.006 \end{pmatrix} \\
&= \begin{pmatrix} 0.21 & -0.1 & -0.086 & -0.024 \\ -0.1 & 0.222 & -0.096 & -0.027 \\ -0.086 & -0.096 & 0.204 & -0.023 \\ -0.024 & -0.027 & -0.023 & 0.074 \end{pmatrix}
\end{aligned}$$

One can verify that each row (and column) in $\widehat{\boldsymbol{\Sigma}}$ sums to zero.

It is worth emphasizing that whereas the modeling of categorical attribute X as a multivariate Bernoulli variable, $\mathbf{X} = (A_1, A_2, \ldots, A_m)^T$, makes the structure of the mean and covariance matrix explicit, the same results would be obtained if we simply treat the mapped values $\mathbf{X}(x_i)$ as a new $n \times m$ binary data matrix, and apply the standard definitions of the mean and covariance matrix from multivariate numeric attribute analysis (see Section 2.3). In essence, the mapping from symbols a_i to binary vectors $\mathbf{e}_i$ is the key idea in categorical attribute analysis.

Example 3.6. Consider the sample $\mathbf{D}$ of size $n = 5$ for the `sepal length` attribute X_1 in the Iris dataset, shown in Table 3.2a. As in Example 3.1, we assume that X_1 has only two categorical values {`Long`, `Short`}. We model X_1 as the multivariate Bernoulli variable $\mathbf{X}_1$ defined as

$$\mathbf{X}_1(v) = \begin{cases} \mathbf{e}_1 = (1,0)^T & \text{if } v = \texttt{Long}(a_1) \\ \mathbf{e}_2 = (0,1)^T & \text{if } v = \texttt{Short}(a_2) \end{cases}$$

The sample mean [Eq. (3.6)] is

$$\hat{\boldsymbol{\mu}} = \hat{\mathbf{p}} = (2/5, 3/5)^T = (0.4, 0.6)^T$$

and the sample covariance matrix [Eq. (3.9)] is

$$\begin{aligned}
\widehat{\boldsymbol{\Sigma}} = \widehat{\mathbf{P}} - \hat{\mathbf{p}}\hat{\mathbf{p}}^T &= \begin{pmatrix} 0.4 & 0 \\ 0 & 0.6 \end{pmatrix} - \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix} \begin{pmatrix} 0.4 & 0.6 \end{pmatrix} \\
&= \begin{pmatrix} 0.4 & 0 \\ 0 & 0.6 \end{pmatrix} - \begin{pmatrix} 0.16 & 0.24 \\ 0.24 & 0.36 \end{pmatrix} = \begin{pmatrix} 0.24 & -0.24 \\ -0.24 & 0.24 \end{pmatrix}
\end{aligned}$$

Table 3.2. (a) Categorical dataset. (b) Mapped binary dataset. (c) Centered dataset.

(a)

	X
x_1	`Short`
x_2	`Short`
x_3	`Long`
x_4	`Short`
x_5	`Long`

(b)

	A_1	A_2
$\mathbf{x}_1$	0	1
$\mathbf{x}_2$	0	1
$\mathbf{x}_3$	1	0
$\mathbf{x}_4$	0	1
$\mathbf{x}_5$	1	0

(c)

	Z_1	Z_2
$\mathbf{z}_1$	−0.4	0.4
$\mathbf{z}_2$	−0.4	0.4
$\mathbf{z}_3$	0.6	−0.6
$\mathbf{z}_4$	−0.4	0.4
$\mathbf{z}_5$	0.6	−0.6

To show that the same result would be obtained via standard numeric analysis, we map the categorical attribute X to the two Bernoulli attributes A_1 and A_2 corresponding to symbols `Long` and `Short`, respectively. The mapped dataset is shown in Table 3.2b. The sample mean is simply

$$\hat{\boldsymbol{\mu}} = \frac{1}{5}\sum_{i=1}^{5} \mathbf{x}_i = \frac{1}{5}(2,3)^T = (0.4, 0.6)^T$$

Next, we center the dataset by subtracting the mean value from each attribute. After centering, the mapped dataset is as shown in Table 3.2c, with attribute Z_i as the centered attribute A_i. We can compute the covariance matrix using the inner-product form [Eq. (2.30)] on the centered column vectors. We have

$$\sigma_1^2 = \frac{1}{5}Z_1^T Z_1 = 1.2/5 = 0.24$$

$$\sigma_2^2 = \frac{1}{5}Z_2^T Z_2 = 1.2/5 = 0.24$$

$$\sigma_{12} = \frac{1}{5}Z_1^T Z_2 = -1.2/5 = -0.24$$

Thus, the sample covariance matrix is given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} 0.24 & -0.24 \\ -0.24 & 0.24 \end{pmatrix}$$

which matches the result obtained by using the multivariate Bernoulli modeling approach.

Multinomial Distribution: Number of Occurrences

Given a multivariate Bernoulli variable $\mathbf{X}$ and a random sample $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ drawn from $\mathbf{X}$. Let N_i be the random variable corresponding to the number of occurrences of symbol a_i in the sample, and let $\mathbf{N} = (N_1, N_2, \ldots, N_m)^T$ denote the vector random variable corresponding to the joint distribution of the number of occurrences over all the symbols. Then $\mathbf{N}$ has a multinomial distribution, given as

$$f\big(\mathbf{N} = (n_1, n_2, \ldots, n_m) \mid \mathbf{p}\big) = \binom{n}{n_1 n_2 \ldots n_m} \prod_{i=1}^{m} p_i^{n_i}$$

We can see that this is a direct generalization of the binomial distribution in Eq. (3.3). The term

$$\binom{n}{n_1 n_2 \dots n_m} = \frac{n!}{n_1! n_2! \dots n_m!}$$

denotes the number of ways of choosing n_i occurrences of each symbol a_i from a sample of size n, with $\sum_{i=1}^{m} n_i = n$.

The mean and covariance matrix of $\mathbf{N}$ are given as n times the mean and covariance matrix of $\mathbf{X}$. That is, the mean of $\mathbf{N}$ is given as

$$\boldsymbol{\mu}_{\mathbf{N}} = E[\mathbf{N}] = nE[\mathbf{X}] = n \cdot \boldsymbol{\mu} = n \cdot \mathbf{p} = \begin{pmatrix} np_1 \\ \vdots \\ np_m \end{pmatrix}$$

and its covariance matrix is given as

$$\boldsymbol{\Sigma}_{\mathbf{N}} = n \cdot (\mathbf{P} - \mathbf{p}\mathbf{p}^T) = \begin{pmatrix} np_1(1-p_1) & -np_1p_2 & \cdots & -np_1p_m \\ -np_1p_2 & np_2(1-p_2) & \cdots & -np_2p_m \\ \vdots & \vdots & \ddots & \vdots \\ -np_1p_m & -np_2p_m & \cdots & np_m(1-p_m) \end{pmatrix}$$

Likewise the sample mean and covariance matrix for $\mathbf{N}$ are given as

$$\hat{\boldsymbol{\mu}}_{\mathbf{N}} = n\hat{\mathbf{p}} \qquad\qquad \widehat{\boldsymbol{\Sigma}}_{\mathbf{N}} = n\big(\widehat{\mathbf{P}} - \hat{\mathbf{p}}\hat{\mathbf{p}}^T\big)$$

3.2 BIVARIATE ANALYSIS

Assume that the data comprises two categorical attributes, X_1 and X_2, with

$$dom(X_1) = \{a_{11}, a_{12}, \dots, a_{1m_1}\}$$
$$dom(X_2) = \{a_{21}, a_{22}, \dots, a_{2m_2}\}$$

We are given n categorical points of the form $\mathbf{x}_i = (x_{i1}, x_{i2})^T$ with $x_{i1} \in dom(X_1)$ and $x_{i2} \in dom(X_2)$. The dataset is thus an $n \times 2$ symbolic data matrix:

$$\mathbf{D} = \begin{pmatrix} X_1 & X_2 \\ \hline x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}$$

We can model X_1 and X_2 as multivariate Bernoulli variables $\mathbf{X}_1$ and $\mathbf{X}_2$ with dimensions m_1 and m_2, respectively. The probability mass functions for $\mathbf{X}_1$ and $\mathbf{X}_2$ are given according to Eq. (3.4):

$$P(\mathbf{X}_1 = \mathbf{e}_{1i}) = f_1(\mathbf{e}_{1i}) = p_i^1 = \prod_{k=1}^{m_1} (p_i^1)^{e_{ik}^1}$$

$$P(\mathbf{X}_2 = \mathbf{e}_{2j}) = f_2(\mathbf{e}_{2j}) = p_j^2 = \prod_{k=1}^{m_2} (p_j^2)^{e_{jk}^2}$$

where $\mathbf{e}_{1i}$ is the ith standard basis vector in $\mathbb{R}^{m_1}$ (for attribute X_1) whose kth component is e_{ik}^1, and $\mathbf{e}_{2j}$ is the jth standard basis vector in $\mathbb{R}^{m_2}$ (for attribute X_2) whose kth component is e_{jk}^2. Further, the parameter p_i^1 denotes the probability of observing symbol a_{1i}, and p_j^2 denotes the probability of observing symbol a_{2j}. Together they must satisfy the conditions: $\sum_{i=1}^{m_1} p_i^1 = 1$ and $\sum_{j=1}^{m_2} p_j^2 = 1$.

The joint distribution of $\mathbf{X}_1$ and $\mathbf{X}_2$ is modeled as the $d' = m_1 + m_2$ dimensional vector variable $\mathbf{X} = \begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix}$, specified by the mapping

$$\mathbf{X}\big((v_1, v_2)^T\big) = \begin{pmatrix}\mathbf{X}_1(v_1)\\ \mathbf{X}_2(v_2)\end{pmatrix} = \begin{pmatrix}\mathbf{e}_{1i}\\ \mathbf{e}_{2j}\end{pmatrix}$$

provided that $v_1 = a_{1i}$ and $v_2 = a_{2j}$. The range of $\mathbf{X}$ thus consists of $m_1 \times m_2$ distinct pairs of vector values $\{(\mathbf{e}_{1i}, \mathbf{e}_{2j})^T\}$, with $1 \le i \le m_1$ and $1 \le j \le m_2$. The joint PMF of $\mathbf{X}$ is given as

$$P\big(\mathbf{X} = (\mathbf{e}_{1i}, \mathbf{e}_{2j})^T\big) = f(\mathbf{e}_{1i}, \mathbf{e}_{2j}) = p_{ij} = \prod_{r=1}^{m_1} \prod_{s=1}^{m_2} p_{ij}^{e_{ir}^1 \cdot e_{js}^2}$$

where p_{ij} the probability of observing the symbol pair (a_{1i}, a_{2j}). These probability parameters must satisfy the condition $\sum_{i=1}^{m_1} \sum_{j=1}^{m_2} p_{ij} = 1$. The joint PMF for $\mathbf{X}$ can be expressed as the $m_1 \times m_2$ matrix

$$\mathbf{P}_{12} = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1m_2} \\ p_{21} & p_{22} & \cdots & p_{2m_2} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m_1 1} & p_{m_1 2} & \cdots & p_{m_1 m_2} \end{pmatrix} \tag{3.10}$$

Example 3.7. Consider the discretized `sepal length` attribute (X_1) in Table 3.1. We also discretize the `sepal width` attribute (X_2) into three values as shown in Table 3.3. We thus have

$$dom(X_1) = \{a_{11} = \texttt{VeryShort}, a_{12} = \texttt{Short}, a_{13} = \texttt{Long}, a_{14} = \texttt{VeryLong}\}$$

$$dom(X_2) = \{a_{21} = \texttt{Short}, a_{22} = \texttt{Medium}, a_{23} = \texttt{Long}\}$$

The symbolic point $\mathbf{x} = (\texttt{Short}, \texttt{Long}) = (a_{12}, a_{23})$, is mapped to the vector

$$\mathbf{X}(\mathbf{x}) = \begin{pmatrix}\mathbf{e}_{12}\\ \mathbf{e}_{23}\end{pmatrix} = (0, 1, 0, 0 \mid 0, 0, 1)^T \in \mathbb{R}^7$$

Table 3.3. Discretized `sepal width` attribute

Bins	Domain	Counts
$[2.0, 2.8]$	Short (a_1)	47
$(2.8, 3.6]$	Medium (a_2)	88
$(3.6, 4.4]$	Long (a_3)	15

where we use $|$ to demarcate the two subvectors $\mathbf{e}_{12} = (0,1,0,0)^T \in \mathbb{R}^4$ and $\mathbf{e}_{23} = (0,0,1)^T \in \mathbb{R}^3$, corresponding to symbolic attributes `sepal length` and `sepal width`, respectively. Note that $\mathbf{e}_{12}$ is the second standard basis vector in $\mathbb{R}^4$ for $\mathbf{X}_1$, and $\mathbf{e}_{23}$ is the third standard basis vector in $\mathbb{R}^3$ for $\mathbf{X}_2$.

Mean

The bivariate mean can easily be generalized from Eq. (3.5), as follows:

$$\boldsymbol{\mu} = E[\mathbf{X}] = E\left[\begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix}\right] = \begin{pmatrix}E[\mathbf{X}_1]\\ E[\mathbf{X}_2]\end{pmatrix} = \begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix} = \begin{pmatrix}\mathbf{p}_1\\ \mathbf{p}_2\end{pmatrix}$$

where $\boldsymbol{\mu}_1 = \mathbf{p}_1 = (p_1^1, \dots, p_{m_1}^1)^T$ and $\boldsymbol{\mu}_2 = \mathbf{p}_2 = (p_1^2, \dots, p_{m_2}^2)^T$ are the mean vectors for $\mathbf{X}_1$ and $\mathbf{X}_2$. The vectors $\mathbf{p}_1$ and $\mathbf{p}_2$ also represent the probability mass functions for $\mathbf{X}_1$ and $\mathbf{X}_2$, respectively.

Sample Mean

The sample mean can also be generalized from Eq. (3.6), by placing a probability mass of $\frac{1}{n}$ at each point:

$$\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i = \frac{1}{n}\begin{pmatrix}\sum_{i=1}^{m_1} n_i^1 \mathbf{e}_{1i}\\ \sum_{j=1}^{m_2} n_j^2 \mathbf{e}_{2j}\end{pmatrix} = \frac{1}{n}\begin{pmatrix}n_1^1\\ \vdots\\ n_{m_1}^1\\ n_1^2\\ \vdots\\ n_{m_2}^2\end{pmatrix} = \begin{pmatrix}\hat{p}_1^1\\ \vdots\\ \hat{p}_{m_1}^1\\ \hat{p}_1^2\\ \vdots\\ \hat{p}_{m_2}^2\end{pmatrix} = \begin{pmatrix}\hat{\mathbf{p}}_1\\ \hat{\mathbf{p}}_2\end{pmatrix} = \begin{pmatrix}\hat{\boldsymbol{\mu}}_1\\ \hat{\boldsymbol{\mu}}_2\end{pmatrix}$$

where n_j^i is the observed frequency of symbol a_{ij} in the sample of size n, and $\hat{\boldsymbol{\mu}}_i = \hat{\mathbf{p}}_i = (p_1^i, p_2^i, \dots, p_{m_i}^i)^T$ is the sample mean vector for $\mathbf{X}_i$, which is also the empirical PMF for attribute $\mathbf{X}_i$.

Covariance Matrix

The covariance matrix for $\mathbf{X}$ is the $d' \times d' = (m_1 + m_2) \times (m_1 + m_2)$ matrix given as

$$\boldsymbol{\Sigma} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{12}^T & \boldsymbol{\Sigma}_{22}\end{pmatrix} \tag{3.11}$$

where $\boldsymbol{\Sigma}_{11}$ is the $m_1 \times m_1$ covariance matrix for $\mathbf{X}_1$, and $\boldsymbol{\Sigma}_{22}$ is the $m_2 \times m_2$ covariance matrix for $\mathbf{X}_2$, which can be computed using Eq. (3.8). That is,

$$\boldsymbol{\Sigma}_{11} = \mathbf{P}_1 - \mathbf{p}_1\mathbf{p}_1^T$$
$$\boldsymbol{\Sigma}_{22} = \mathbf{P}_2 - \mathbf{p}_2\mathbf{p}_2^T$$

where $\mathbf{P}_1 = \text{diag}(\mathbf{p}_1)$ and $\mathbf{P}_2 = \text{diag}(\mathbf{p}_2)$. Further, $\boldsymbol{\Sigma}_{12}$ is the $m_1 \times m_2$ covariance matrix between variables $\mathbf{X}_1$ and $\mathbf{X}_2$, given as

$$\begin{aligned}
\boldsymbol{\Sigma}_{12} &= E[(\mathbf{X}_1 - \boldsymbol{\mu}_1)(\mathbf{X}_2 - \boldsymbol{\mu}_2)^T] \\
&= E[\mathbf{X}_1\mathbf{X}_2^T] - E[\mathbf{X}_1]E[\mathbf{X}_2]^T \\
&= \mathbf{P}_{12} - \boldsymbol{\mu}_1\boldsymbol{\mu}_2^T \\
&= \mathbf{P}_{12} - \mathbf{p}_1\mathbf{p}_2^T \\
&= \begin{pmatrix}
p_{11} - p_1^1 p_1^2 & p_{12} - p_1^1 p_2^2 & \cdots & p_{1m_2} - p_1^1 p_{m_2}^2 \\
p_{21} - p_2^1 p_1^2 & p_{22} - p_2^1 p_2^2 & \cdots & p_{2m_2} - p_2^1 p_{m_2}^2 \\
\vdots & \vdots & \ddots & \vdots \\
p_{m_1 1} - p_{m_1}^1 p_1^2 & p_{m_1 2} - p_{m_1}^1 p_2^2 & \cdots & p_{m_1 m_2} - p_{m_1}^1 p_{m_2}^2
\end{pmatrix}
\end{aligned}$$

where $\mathbf{P}_{12}$ represents the joint PMF for $\mathbf{X}$ given in Eq. (3.10).

Incidentally, each row and each column of $\boldsymbol{\Sigma}_{12}$ sums to zero. For example, consider row i and column j:

$$\sum_{k=1}^{m_2}(p_{ik} - p_i^1 p_k^2) = \left(\sum_{k=1}^{m_2} p_{ik}\right) - p_i^1 = p_i^1 - p_i^1 = 0$$

$$\sum_{k=1}^{m_1}(p_{kj} - p_k^1 p_j^2) = \left(\sum_{k=1}^{m_1} p_{kj}\right) - p_j^2 = p_j^2 - p_j^2 = 0$$

which follows from the fact that summing the joint mass function over all values of $\mathbf{X}_2$, yields the marginal distribution of $\mathbf{X}_1$, and summing it over all values of $\mathbf{X}_1$ yields the marginal distribution for $\mathbf{X}_2$. Note that p_j^2 is the probability of observing symbol a_{2j}; it should not be confused with the square of p_j. Combined with the fact that $\boldsymbol{\Sigma}_{11}$ and $\boldsymbol{\Sigma}_{22}$ also have row and column sums equal to zero via Eq. (3.7), the full covariance matrix $\boldsymbol{\Sigma}$ has rows and columns that sum up to zero.

Sample Covariance Matrix

The sample covariance matrix is given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} \widehat{\boldsymbol{\Sigma}}_{11} & \widehat{\boldsymbol{\Sigma}}_{12} \\ \widehat{\boldsymbol{\Sigma}}_{12}^T & \widehat{\boldsymbol{\Sigma}}_{22} \end{pmatrix} \tag{3.12}$$

where

$$\begin{aligned}
\widehat{\boldsymbol{\Sigma}}_{11} &= \widehat{\mathbf{P}}_1 - \hat{\mathbf{p}}_1\hat{\mathbf{p}}_1^T \\
\widehat{\boldsymbol{\Sigma}}_{22} &= \widehat{\mathbf{P}}_2 - \hat{\mathbf{p}}_2\hat{\mathbf{p}}_2^T \\
\widehat{\boldsymbol{\Sigma}}_{12} &= \widehat{\mathbf{P}}_{12} - \hat{\mathbf{p}}_1\hat{\mathbf{p}}_2^T
\end{aligned}$$

Here $\widehat{\mathbf{P}}_1 = \text{diag}(\hat{\mathbf{p}}_1)$ and $\widehat{\mathbf{P}}_2 = \text{diag}(\hat{\mathbf{p}}_2)$, and $\hat{\mathbf{p}}_1$ and $\hat{\mathbf{p}}_2$ specify the empirical probability mass functions for $\mathbf{X}_1$, and $\mathbf{X}_2$, respectively. Further, $\widehat{\mathbf{P}}_{12}$ specifies the empirical joint PMF for $\mathbf{X}_1$ and $\mathbf{X}_2$, given as

$$\widehat{\mathbf{P}}_{12}(i,j) = \hat{f}(\mathbf{e}_{1i}, \mathbf{e}_{2j}) = \frac{1}{n}\sum_{k=1}^{n} I_{ij}(\mathbf{x}_k) = \frac{n_{ij}}{n} = \hat{p}_{ij} \tag{3.13}$$

where I_{ij} is the indicator variable

$$I_{ij}(\mathbf{x}_k) = \begin{cases} 1 & \text{if } \mathbf{x}_{k1} = \mathbf{e}_{1i} \text{ and } \mathbf{x}_{k2} = \mathbf{e}_{2j} \\ 0 & \text{otherwise} \end{cases}$$

Taking the sum of $I_{ij}(\mathbf{x}_k)$ over all the n points in the sample yields the number of occurrences, n_{ij}, of the symbol pair (a_{1i}, a_{2j}) in the sample. One issue with the cross-attribute covariance matrix $\widehat{\mathbf{\Sigma}}_{12}$ is the need to estimate a quadratic number of parameters. That is, we need to obtain reliable counts n_{ij} to estimate the parameters p_{ij}, for a total of $O(m_1 \times m_2)$ parameters that have to be estimated, which can be a problem if the categorical attributes have many symbols. On the other hand, estimating $\widehat{\mathbf{\Sigma}}_{11}$ and $\widehat{\mathbf{\Sigma}}_{22}$ requires that we estimate m_1 and m_2 parameters, corresponding to p_i^1 and p_j^2, respectively. In total, computing $\mathbf{\Sigma}$ requires the estimation of $m_1m_2 + m_1 + m_2$ parameters.

Example 3.8. We continue with the bivariate categorical attributes X_1 and X_2 in Example 3.7. From Example 3.4, and from the occurrence counts for each of the values of `sepal width` in Table 3.3, we have

$$\hat{\boldsymbol{\mu}}_1 = \hat{\mathbf{p}}_1 = \begin{pmatrix} 0.3 \\ 0.333 \\ 0.287 \\ 0.08 \end{pmatrix} \qquad \hat{\boldsymbol{\mu}}_2 = \hat{\mathbf{p}}_2 = \frac{1}{150}\begin{pmatrix} 47 \\ 88 \\ 15 \end{pmatrix} = \begin{pmatrix} 0.313 \\ 0.587 \\ 0.1 \end{pmatrix}$$

Thus, the mean for $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}$ is given as

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\boldsymbol{\mu}}_1 \\ \hat{\boldsymbol{\mu}}_2 \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{p}}_1 \\ \hat{\mathbf{p}}_2 \end{pmatrix} = (0.3, 0.333, 0.287, 0.08 \,|\, 0.313, 0.587, 0.1)^T$$

From Example 3.5 we have

$$\widehat{\mathbf{\Sigma}}_{11} = \begin{pmatrix} 0.21 & -0.1 & -0.086 & -0.024 \\ -0.1 & 0.222 & -0.096 & -0.027 \\ -0.086 & -0.096 & 0.204 & -0.023 \\ -0.024 & -0.027 & -0.023 & 0.074 \end{pmatrix}$$

In a similar manner we can obtain

$$\widehat{\mathbf{\Sigma}}_{22} = \begin{pmatrix} 0.215 & -0.184 & -0.031 \\ -0.184 & 0.242 & -0.059 \\ -0.031 & -0.059 & 0.09 \end{pmatrix}$$

Next, we use the observed counts in Table 3.4 to obtain the empirical joint PMF for $\mathbf{X}_1$ and $\mathbf{X}_2$ using Eq. (3.13), as plotted in Figure 3.2. From these probabilities we get

$$E[\mathbf{X}_1\mathbf{X}_2^T] = \widehat{\mathbf{P}}_{12} = \frac{1}{150}\begin{pmatrix} 7 & 33 & 5 \\ 24 & 18 & 8 \\ 13 & 30 & 0 \\ 3 & 7 & 2 \end{pmatrix} = \begin{pmatrix} 0.047 & 0.22 & 0.033 \\ 0.16 & 0.12 & 0.053 \\ 0.087 & 0.2 & 0 \\ 0.02 & 0.047 & 0.013 \end{pmatrix}$$

Table 3.4. Observed Counts (n_{ij}): `sepal length` and `sepal width`

		X_2		
		`Short` ($\mathbf{e}_{21}$)	`Medium` ($\mathbf{e}_{22}$)	`Long` ($\mathbf{e}_{23}$)
X_1	`Very Short` ($\mathbf{e}_{11}$)	7	33	5
	`Short` ($\mathbf{e}_{22}$)	24	18	8
	`Long` ($\mathbf{e}_{13}$)	13	30	0
	`Very Long` ($\mathbf{e}_{14}$)	3	7	2

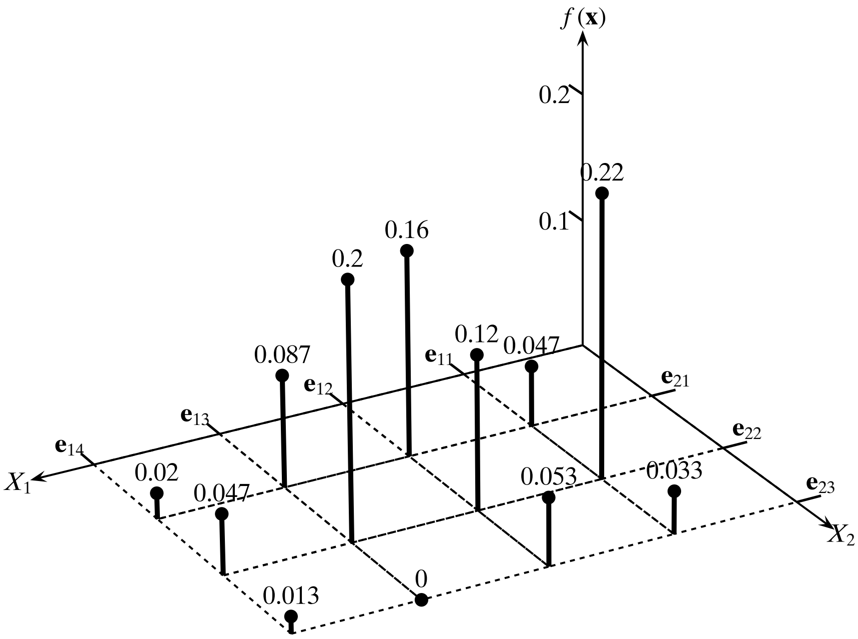

Figure 3.2. Empirical joint probability mass function: `sepal length` and `sepal width`.

Further, we have

$$
\begin{aligned}
E[\mathbf{X}_1]E[\mathbf{X}_2]^T &= \hat{\boldsymbol{\mu}}_1\hat{\boldsymbol{\mu}}_2^T = \hat{\mathbf{p}}_1\hat{\mathbf{p}}_2^T \\
&= \begin{pmatrix} 0.3 \\ 0.333 \\ 0.287 \\ 0.08 \end{pmatrix} \begin{pmatrix} 0.313 & 0.587 & 0.1 \end{pmatrix} \\
&= \begin{pmatrix} 0.094 & 0.176 & 0.03 \\ 0.104 & 0.196 & 0.033 \\ 0.09 & 0.168 & 0.029 \\ 0.025 & 0.047 & 0.008 \end{pmatrix}
\end{aligned}
$$

We can now compute the across-attribute sample covariance matrix $\widehat{\mathbf{\Sigma}}_{12}$ for $\mathbf{X}_1$ and $\mathbf{X}_2$ using Eq. (3.11), as follows:

$$\widehat{\mathbf{\Sigma}}_{12} = \widehat{\mathbf{P}}_{12} - \hat{\mathbf{p}}_1 \hat{\mathbf{p}}_2^T = \begin{pmatrix} -0.047 & 0.044 & 0.003 \\ 0.056 & -0.076 & 0.02 \\ -0.003 & 0.032 & -0.029 \\ -0.005 & 0 & 0.005 \end{pmatrix}$$

One can observe that each row and column in $\widehat{\mathbf{\Sigma}}_{12}$ sums to zero. Putting it all together, from $\widehat{\mathbf{\Sigma}}_{11}$, $\widehat{\mathbf{\Sigma}}_{22}$ and $\widehat{\mathbf{\Sigma}}_{12}$ we obtain the sample covariance matrix as follows

$$\widehat{\mathbf{\Sigma}} = \begin{pmatrix} \widehat{\mathbf{\Sigma}}_{11} & \widehat{\mathbf{\Sigma}}_{12} \\ \widehat{\mathbf{\Sigma}}_{12}^T & \widehat{\mathbf{\Sigma}}_{22} \end{pmatrix} = \left(\begin{array}{rrrr|rrr} 0.21 & -0.1 & -0.086 & -0.024 & -0.047 & 0.044 & 0.003 \\ -0.1 & 0.222 & -0.096 & -0.027 & 0.056 & -0.076 & 0.02 \\ -0.086 & -0.096 & 0.204 & -0.023 & -0.003 & 0.032 & -0.029 \\ -0.024 & -0.027 & -0.023 & 0.074 & -0.005 & 0 & 0.005 \\ \hline -0.047 & 0.056 & -0.003 & -0.005 & 0.215 & -0.184 & -0.031 \\ 0.044 & -0.076 & 0.032 & 0 & -0.184 & 0.242 & -0.059 \\ 0.003 & 0.02 & -0.029 & 0.005 & -0.031 & -0.059 & 0.09 \end{array}\right)$$

In $\widehat{\mathbf{\Sigma}}$, each row and column also sums to zero.

3.2.1 Attribute Dependence: Contingency Analysis

Testing for the independence of the two categorical random variables X_1 and X_2 can be done via *contingency table analysis*. The main idea is to set up a hypothesis testing framework, where the null hypothesis H_0 is that $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent, and the alternative hypothesis H_1 is that they are dependent. We then compute the value of the chi-square statistic χ^2 under the null hypothesis. Depending on the p-value, we either accept or reject the null hypothesis; in the latter case the attributes are considered to be dependent.

Contingency Table

A contingency table for $\mathbf{X}_1$ and $\mathbf{X}_2$ is the $m_1 \times m_2$ matrix of observed counts n_{ij} for all pairs of values $(\mathbf{e}_{1i}, \mathbf{e}_{2j})$ in the given sample of size n, defined as

$$\mathbf{N}_{12} = n \cdot \widehat{\mathbf{P}}_{12} = \begin{pmatrix} n_{11} & n_{12} & \cdots & n_{1m_2} \\ n_{21} & n_{22} & \cdots & n_{2m_2} \\ \vdots & \vdots & \ddots & \vdots \\ n_{m_1 1} & n_{m_1 2} & \cdots & n_{m_1 m_2} \end{pmatrix}$$

Table 3.5. Contingency table: `sepal length` vs. `sepal width`

Sepal length (X_1)		Sepal width (X_2)			
		Short a_{21}	Medium a_{22}	Long a_{23}	Row Counts
	Very Short (a_{11})	7	33	5	$n_1^1 = 45$
	Short (a_{12})	24	18	8	$n_2^1 = 50$
	Long (a_{13})	13	30	0	$n_3^1 = 43$
	Very Long (a_{14})	3	7	2	$n_4^1 = 12$
	Column Counts	$n_1^2 = 47$	$n_2^2 = 88$	$n_3^2 = 15$	$n = 150$

where $\widehat{\mathbf{P}}_{12}$ is the empirical joint PMF for $\mathbf{X}_1$ and $\mathbf{X}_2$, computed via Eq. (3.13). The contingency table is then augmented with row and column marginal counts, as follows:

$$\mathbf{N}_1 = n \cdot \hat{\mathbf{p}}_1 = \begin{pmatrix} n_1^1 \\ \vdots \\ n_{m_1}^1 \end{pmatrix} \qquad \mathbf{N}_2 = n \cdot \hat{\mathbf{p}}_2 = \begin{pmatrix} n_1^2 \\ \vdots \\ n_{m_2}^2 \end{pmatrix}$$

Note that the marginal row and column entries and the sample size satisfy the following constraints:

$$n_i^1 = \sum_{j=1}^{m_2} n_{ij} \qquad n_j^2 = \sum_{i=1}^{m_1} n_{ij} \qquad n = \sum_{i=1}^{m_1} n_i^1 = \sum_{j=1}^{m_2} n_j^2 = \sum_{i=1}^{m_1}\sum_{j=1}^{m_2} n_{ij}$$

It is worth noting that both $\mathbf{N}_1$ and $\mathbf{N}_2$ have a multinomial distribution with parameters $\mathbf{p}_1 = (p_1^1, \ldots, p_{m_1}^1)$ and $\mathbf{p}_2 = (p_1^2, \ldots, p_{m_2}^2)$, respectively. Further, $\mathbf{N}_{12}$ also has a multinomial distribution with parameters $\mathbf{P}_{12} = \{p_{ij}\}$, for $1 \le i \le m_1$ and $1 \le j \le m_2$.

Example 3.9 (Contingency Table). Table 3.4 shows the observed counts for the discretized `sepal length` (X_1) and `sepal width` (X_2) attributes. Augmenting the table with the row and column marginal counts and the sample size yields the final contingency table shown in Table 3.5.

χ^2 Statistic and Hypothesis Testing

Under the null hypothesis $\mathbf{X}_1$ and $\mathbf{X}_2$ are assumed to be independent, which means that their joint probability mass function is given as

$$\hat{p}_{ij} = \hat{p}_i^1 \cdot \hat{p}_j^2$$

Under this independence assumption, the expected frequency for each pair of values is given as

$$e_{ij} = n \cdot \hat{p}_{ij} = n \cdot \hat{p}_i^1 \cdot \hat{p}_j^2 = n \cdot \frac{n_i^1}{n} \cdot \frac{n_j^2}{n} = \frac{n_i^1 n_j^2}{n} \tag{3.14}$$

However, from the sample we already have the observed frequency of each pair of values, n_{ij}. We would like to determine whether there is a significant difference in the observed and expected frequencies for each pair of values. If there is no significant difference, then the independence assumption is valid and we accept the null hypothesis that the attributes are independent. On the other hand, if there is a significant difference, then the null hypothesis should be rejected and we conclude that the attributes are dependent.

The χ^2 statistic quantifies the difference between observed and expected counts for each pair of values; it is defined as follows:

$$\chi^2 = \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{(n_{ij} - e_{ij})^2}{e_{ij}} \tag{3.15}$$

At this point, we need to determine the probability of obtaining the computed χ^2 value. In general, this can be rather difficult if we do not know the sampling distribution of a given statistic. Fortunately, for the χ^2 statistic it is known that its sampling distribution follows the *chi-squared* density function with q degrees of freedom:

$$f(x|q) = \frac{1}{2^{q/2}\Gamma(q/2)} x^{\frac{q}{2}-1} e^{-\frac{x}{2}} \tag{3.16}$$

where the gamma function Γ is defined as

$$\Gamma(k > 0) = \int_0^\infty x^{k-1} e^{-x} dx \tag{3.17}$$

The degrees of freedom, q, represent the number of independent parameters. In the contingency table there are $m_1 \times m_2$ observed counts n_{ij}. However, note that each row i and each column j must sum to n_i^1 and n_j^2, respectively. Further, the sum of the row and column marginals must also add to n; thus we have to remove $(m_1 + m_2)$ parameters from the number of independent parameters. However, doing this removes one of the parameters, say $n_{m_1 m_2}$, twice, so we have to add back one to the count. The total degrees of freedom is therefore

$$\begin{aligned} q &= |dom(X_1)| \times |dom(X_2)| - (|dom(X_1)| + |dom(X_2)|) + 1 \\ &= m_1 m_2 - m_1 - m_2 + 1 \\ &= (m_1 - 1)(m_2 - 1) \end{aligned}$$

p-value

The *p-value* of a statistic θ is defined as the probability of obtaining a value at least as extreme as the observed value, say z, under the null hypothesis, defined as

$$p\text{-value}(z) = P(\theta \ge z) = 1 - F(\theta)$$

where $F(\theta)$ is the cumulative probability distribution for the statistic.

The p-value gives a measure of how surprising is the observed value of the statistic. If the observed value lies in a low-probability region, then the value is more surprising. In general, the lower the p-value, the more surprising the observed value, and the

Table 3.6. Expected counts

		X_2		
		`Short` (a_{21})	`Medium` (a_{22})	`Long` (a_{23})
X_1	`Very Short` (a_{11})	14.1	26.4	4.5
	`Short` (a_{12})	15.67	29.33	5.0
	`Long` (a_{13})	13.47	25.23	4.3
	`Very Long` (a_{14})	3.76	7.04	1.2

more the grounds for rejecting the null hypothesis. The null hypothesis is rejected if the p-value is below some *significance level*, α. For example, if $\alpha = 0.01$, then we reject the null hypothesis if $p\text{-value}(z) \leq \alpha$. The significance level α corresponds to the probability of rejecting the null hypothesis when it is true. For a given significance level α, the value of the test statistic, say z, with a p-value of $p\text{-value}(z) = \alpha$, is called a *critical value*. An alternative test for rejection of the null hypothesis is to check if $\chi^2 > z$, as in that case the p-value of the observed χ^2 value is bounded by α, that is, $p\text{-value}(\chi^2) \leq p\text{-value}(z) = \alpha$. The value $1 - \alpha$ is also called the *confidence level*.

Example 3.10. Consider the contingency table for `sepal length` and `sepal width` in Table 3.5. We compute the expected counts using Eq. (3.14); these counts are shown in Table 3.6. For example, we have

$$e_{11} = \frac{n_1^1 n_1^2}{n} = \frac{45 \cdot 47}{150} = \frac{2115}{150} = 14.1$$

Next we use Eq. (3.15) to compute the value of the χ^2 statistic, which is given as $\chi^2 = 21.8$.

Further, the number of degrees of freedom is given as

$$q = (m_1 - 1) \cdot (m_2 - 1) = 3 \cdot 2 = 6$$

The plot of the chi-squared density function with 6 degrees of freedom is shown in Figure 3.3. From the cumulative chi-squared distribution, we obtain

$$p\text{-value}(21.8) = 1 - F(21.8|6) = 1 - 0.9987 = 0.0013$$

At a significance level of $\alpha = 0.01$, we would certainly be justified in rejecting the null hypothesis because the large value of the χ^2 statistic is indeed surprising. Further, at the 0.01 significance level, the critical value of the statistic is

$$z = F^{-1}(1 - 0.01|6) = F^{-1}(0.99|6) = 16.81$$

This critical value is also shown in Figure 3.3, and we can clearly see that the observed value of 21.8 is in the rejection region, as $21.8 > z = 16.81$. In effect, we reject the null hypothesis that `sepal length` and `sepal width` are independent, and accept the alternative hypothesis that they are dependent.

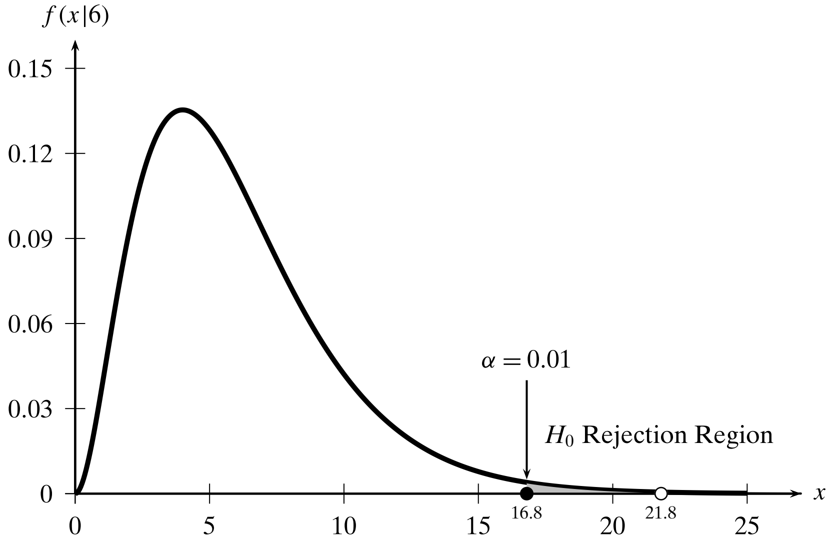

Figure 3.3. Chi-squared distribution ($q = 6$).

3.3 MULTIVARIATE ANALYSIS

Assume that the dataset comprises d categorical attributes X_j $(1 \le j \le d)$ with $dom(X_j) = \{a_{j1}, a_{j2}, \dots, a_{jm_j}\}$. We are given n categorical points of the form $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{id})^T$ with $x_{ij} \in dom(X_j)$. The dataset is thus an $n \times d$ symbolic matrix

$$\mathbf{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_d \\ \hline x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}$$

Each attribute X_i is modeled as an m_i-dimensional multivariate Bernoulli variable $\mathbf{X}_i$, and their joint distribution is modeled as a $d' = \sum_{j=1}^{d} m_j$ dimensional vector random variable

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \vdots \\ \mathbf{X}_d \end{pmatrix}$$

Each categorical data point $\mathbf{v} = (v_1, v_2, \dots, v_d)^T$ is therefore represented as a d'-dimensional binary vector

$$\mathbf{X}(\mathbf{v}) = \begin{pmatrix} \mathbf{X}_1(v_1) \\ \vdots \\ \mathbf{X}_d(v_d) \end{pmatrix} = \begin{pmatrix} \mathbf{e}_{1k_1} \\ \vdots \\ \mathbf{e}_{dk_d} \end{pmatrix}$$

provided $v_i = a_{ik_i}$, the k_ith symbol of X_i. Here $\mathbf{e}_{ik_i}$ is the k_ith standard basis vector in $\mathbb{R}^{m_i}$.

Mean

Generalizing from the bivariate case, the mean and sample mean for $\mathbf{X}$ are given as

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \vdots \\ \boldsymbol{\mu}_d \end{pmatrix} = \begin{pmatrix} \mathbf{p}_1 \\ \vdots \\ \mathbf{p}_d \end{pmatrix} \qquad\qquad \hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\boldsymbol{\mu}}_1 \\ \vdots \\ \hat{\boldsymbol{\mu}}_d \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{p}}_1 \\ \vdots \\ \hat{\mathbf{p}}_d \end{pmatrix}$$

where $\mathbf{p}_i = (p_1^i, \ldots, p_{m_i}^i)^T$ is the PMF for $\mathbf{X}_i$, and $\hat{\mathbf{p}}_i = (\hat{p}_1^i, \ldots, \hat{p}_{m_i}^i)^T$ is the empirical PMF for $\mathbf{X}_i$.

Covariance Matrix

The covariance matrix for $\mathbf{X}$, and its estimate from the sample, are given as the $d' \times d'$ matrices:

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} & \cdots & \boldsymbol{\Sigma}_{1d} \\ \boldsymbol{\Sigma}_{12}^T & \boldsymbol{\Sigma}_{22} & \cdots & \boldsymbol{\Sigma}_{2d} \\ \cdots & \cdots & \ddots & \cdots \\ \boldsymbol{\Sigma}_{1d}^T & \boldsymbol{\Sigma}_{2d}^T & \cdots & \boldsymbol{\Sigma}_{dd} \end{pmatrix} \qquad\qquad \widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} \widehat{\boldsymbol{\Sigma}}_{11} & \widehat{\boldsymbol{\Sigma}}_{12} & \cdots & \widehat{\boldsymbol{\Sigma}}_{1d} \\ \widehat{\boldsymbol{\Sigma}}_{12}^T & \widehat{\boldsymbol{\Sigma}}_{22} & \cdots & \widehat{\boldsymbol{\Sigma}}_{2d} \\ \cdots & \cdots & \ddots & \cdots \\ \widehat{\boldsymbol{\Sigma}}_{1d}^T & \widehat{\boldsymbol{\Sigma}}_{2d}^T & \cdots & \widehat{\boldsymbol{\Sigma}}_{dd} \end{pmatrix}$$

where $d' = \sum_{i=1}^d m_i$, and $\boldsymbol{\Sigma}_{ij}$ (and $\widehat{\boldsymbol{\Sigma}}_{ij}$) is the $m_i \times m_j$ covariance matrix (and its estimate) for attributes $\mathbf{X}_i$ and $\mathbf{X}_j$:

$$\boldsymbol{\Sigma}_{ij} = \mathbf{P}_{ij} - \mathbf{p}_i \mathbf{p}_j^T \qquad\qquad \widehat{\boldsymbol{\Sigma}}_{ij} = \widehat{\mathbf{P}}_{ij} - \hat{\mathbf{p}}_i \hat{\mathbf{p}}_j^T \tag{3.18}$$

Here $\mathbf{P}_{ij}$ is the joint PMF and $\widehat{\mathbf{P}}_{ij}$ is the empirical joint PMF for $\mathbf{X}_i$ and $\mathbf{X}_j$, which can be computed using Eq. (3.13).

Example 3.11 (Multivariate Analysis). Let us consider the 3-dimensional subset of the Iris dataset, with the discretized attributes `sepal length` (X_1) and `sepal width` (X_2), and the categorical attribute `class` (X_3). The domains for X_1 and X_2 are given in Table 3.1 and Table 3.3, respectively, and $dom(X_3) =$ {`iris-versicolor`, `iris-setosa`, `iris-virginica`}. Each value of X_3 occurs 50 times.

The categorical point $\mathbf{x} =$ (`Short`, `Medium`, `iris-versicolor`) is modeled as the vector

$$\mathbf{X}(\mathbf{x}) = \begin{pmatrix} \mathbf{e}_{12} \\ \mathbf{e}_{22} \\ \mathbf{e}_{31} \end{pmatrix} = (0,1,0,0 \mid 0,1,0 \mid 1,0,0)^T \in \mathbb{R}^{10}$$

From Example 3.8 and the fact that each value in $dom(X_3)$ occurs 50 times in a sample of $n = 150$, the sample mean is given as

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\boldsymbol{\mu}}_1 \\ \hat{\boldsymbol{\mu}}_2 \\ \hat{\boldsymbol{\mu}}_3 \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{p}}_1 \\ \hat{\mathbf{p}}_2 \\ \hat{\mathbf{p}}_3 \end{pmatrix} = (0.3, 0.333, 0.287, 0.08 \mid 0.313, 0.587, 0.1 \mid 0.33, 0.33, 0.33)^T$$

Using $\hat{\mathbf{p}}_3 = (0.33, 0.33, 0.33)^T$ we can compute the sample covariance matrix for X_3 using Eq. (3.9):

$$\widehat{\boldsymbol{\Sigma}}_{33} = \begin{pmatrix} 0.222 & -0.111 & -0.111 \\ -0.111 & 0.222 & -0.111 \\ -0.111 & -0.111 & 0.222 \end{pmatrix}$$

Using Eq. (3.18) we obtain

$$\widehat{\boldsymbol{\Sigma}}_{13} = \begin{pmatrix} -0.067 & 0.16 & -0.093 \\ 0.082 & -0.038 & -0.044 \\ 0.011 & -0.096 & 0.084 \\ -0.027 & -0.027 & 0.053 \end{pmatrix}$$

$$\widehat{\boldsymbol{\Sigma}}_{23} = \begin{pmatrix} 0.076 & -0.098 & 0.022 \\ -0.042 & 0.044 & -0.002 \\ -0.033 & 0.053 & -0.02 \end{pmatrix}$$

Combined with $\widehat{\boldsymbol{\Sigma}}_{11}$, $\widehat{\boldsymbol{\Sigma}}_{22}$ and $\widehat{\boldsymbol{\Sigma}}_{12}$ from Example 3.8, the final sample covariance matrix is the 10×10 symmetric matrix given as

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} \widehat{\boldsymbol{\Sigma}}_{11} & \widehat{\boldsymbol{\Sigma}}_{12} & \widehat{\boldsymbol{\Sigma}}_{13} \\ \widehat{\boldsymbol{\Sigma}}_{12}^T & \widehat{\boldsymbol{\Sigma}}_{22} & \widehat{\boldsymbol{\Sigma}}_{23} \\ \widehat{\boldsymbol{\Sigma}}_{13}^T & \widehat{\boldsymbol{\Sigma}}_{23}^T & \widehat{\boldsymbol{\Sigma}}_{33} \end{pmatrix}$$

3.3.1 Multiway Contingency Analysis

For multiway dependence analysis, we have to first determine the empirical joint probability mass function for $\mathbf{X}$:

$$\hat{f}(\mathbf{e}_{1i_1}, \mathbf{e}_{2i_2}, \ldots, \mathbf{e}_{di_d}) = \frac{1}{n}\sum_{k=1}^{n} I_{i_1 i_2 \ldots i_d}(\mathbf{x}_k) = \frac{n_{i_1 i_2 \ldots i_d}}{n} = \hat{p}_{i_1 i_2 \ldots i_d}$$

where $I_{i_1 i_2 \ldots i_d}$ is the indicator variable

$$I_{i_1 i_2 \ldots i_d}(\mathbf{x}_k) = \begin{cases} 1 & \text{if } x_{k1} = \mathbf{e}_{1i_1}, x_{k2} = \mathbf{e}_{2i_2}, \ldots, x_{kd} = \mathbf{e}_{di_d} \\ 0 & \text{otherwise} \end{cases}$$

The sum of $I_{i_1 i_2 \ldots i_d}$ over all the n points in the sample yields the number of occurrences, $n_{i_1 i_2 \ldots i_d}$, of the symbolic vector $(a_{1i_1}, a_{2i_2}, \ldots, a_{di_d})$. Dividing the occurrences by the sample size results in the probability of observing those symbols. Using the notation $\mathbf{i} = (i_1, i_2, \ldots, i_d)$ to denote the index tuple, we can write the joint empirical PMF as the d-dimensional matrix $\widehat{\mathbf{P}}$ of size $m_1 \times m_2 \times \cdots \times m_d = \prod_{i=1}^{d} m_i$, given as

$$\widehat{\mathbf{P}}(\mathbf{i}) = \{\hat{p}_{\mathbf{i}}\} \text{ for all index tuples } \mathbf{i}, \text{ with } 1 \le i_1 \le m_1, \ldots, 1 \le i_d \le m_d$$

where $\hat{p}_{\mathbf{i}} = \hat{p}_{i_1 i_2 \ldots i_d}$. The d-dimensional contingency table is then given as

$$\mathbf{N} = n \times \widehat{\mathbf{P}} = \{n_{\mathbf{i}}\} \text{ for all index tuples } \mathbf{i}, \text{ with } 1 \le i_1 \le m_1, \ldots, 1 \le i_d \le m_d$$

where $n_{\mathbf{i}} = n_{i_1 i_2 \ldots i_d}$. The contingency table is augmented with the marginal count vectors $\mathbf{N}_i$ for all d attributes $\mathbf{X}_i$:

$$\mathbf{N}_i = n\hat{\mathbf{p}}_i = \begin{pmatrix} n^i_1 \\ \vdots \\ n^i_{m_i} \end{pmatrix}$$

where $\hat{\mathbf{p}}_i$ is the empirical PMF for $\mathbf{X}_i$.

χ^2-Test

We can test for a d-way dependence between the d categorical attributes using the null hypothesis H_0 that they are d-way independent. The alternative hypothesis H_1 is that they are not d-way independent, that is, they are dependent in some way. Note that d-dimensional contingency analysis indicates whether all d attributes taken together are independent or not. In general we may have to conduct k-way contingency analysis to test if any subset of $k \leq d$ attributes are independent or not.

Under the null hypothesis, the expected number of occurrences of the symbol tuple $(a_{1i_1}, a_{2i_2}, \ldots, a_{di_d})$ is given as

$$e_{\mathbf{i}} = n \cdot \hat{p}_{\mathbf{i}} = n \cdot \prod_{j=1}^{d} \hat{p}^j_{i_j} = \frac{n^1_{i_1} n^2_{i_2} \ldots n^d_{i_d}}{n^{d-1}} \tag{3.19}$$

The chi-squared statistic measures the difference between the observed counts $n_{\mathbf{i}}$ and the expected counts $e_{\mathbf{i}}$:

$$\chi^2 = \sum_{\mathbf{i}} \frac{(n_{\mathbf{i}} - e_{\mathbf{i}})^2}{e_{\mathbf{i}}} = \sum_{i_1=1}^{m_1} \sum_{i_2=1}^{m_2} \cdots \sum_{i_d=1}^{m_d} \frac{(n_{i_1,i_2,\ldots,i_d} - e_{i_1,i_2,\ldots,i_d})^2}{e_{i_1,i_2,\ldots,i_d}} \tag{3.20}$$

The χ^2 statistic follows a chi-squared density function with q degrees of freedom. For the d-way contingency table we can compute q by noting that there are ostensibly $\prod_{i=1}^{d} |dom(X_i)|$ independent parameters (the counts). However, we have to remove $\sum_{i=1}^{d} |dom(X_i)|$ degrees of freedom because the marginal count vector along each dimension $\mathbf{X}_i$ must equal $\mathbf{N}_i$. However, doing so removes one of the parameters d times, so we need to add back $d-1$ to the free parameters count. The total number of degrees of freedom is given as

$$\begin{aligned} q &= \prod_{i=1}^{d} |dom(X_i)| - \sum_{i=1}^{d} |dom(X_i)| + (d-1) \\ &= \Big(\prod_{i=1}^{d} m_i\Big) - \Big(\sum_{i=1}^{d} m_i\Big) + d - 1 \end{aligned} \tag{3.21}$$

To reject the null hypothesis, we have to check whether the p-value of the observed χ^2 value is smaller than the desired significance level α (say $\alpha = 0.01$) using the chi-squared density with q degrees of freedom [Eq. (3.16)].

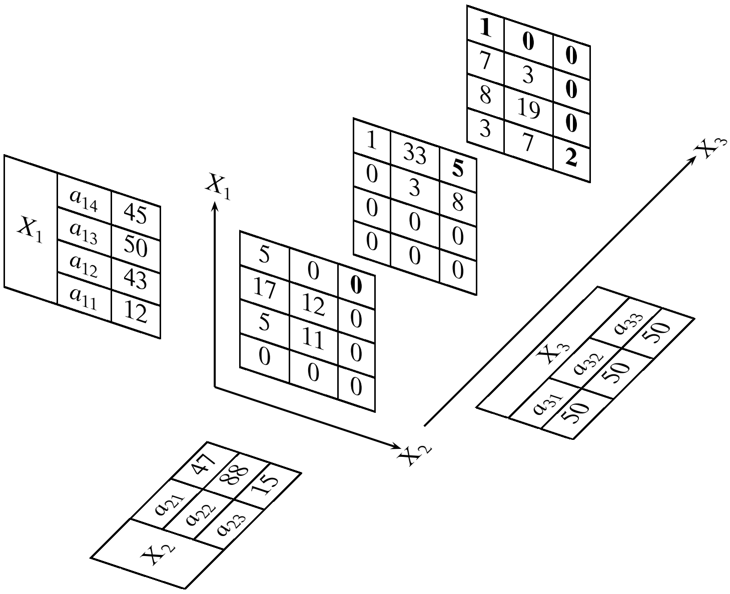

Figure 3.4. 3-Way contingency table (with marginal counts along each dimension).

Table 3.7. 3-Way expected counts

		$X_3(a_{31}/a_{32}/a_{33})$		
		X_2		
		a_{21}	a_{22}	a_{23}
X_1	a_{11}	1.25	2.35	0.40
	a_{12}	4.49	8.41	1.43
	a_{13}	5.22	9.78	1.67
	a_{14}	4.70	8.80	1.50

Example 3.12. Consider the 3-way contingency table in Figure 3.4. It shows the observed counts for each tuple of symbols (a_{1i}, a_{2j}, a_{3k}) for the three attributes `sepal length` (X_1), `sepal width` (X_2), and `class` (X_3). From the marginal counts for X_1 and X_2 in Table 3.5, and the fact that all three values of X_3 occur 50 times, we can compute the expected counts [Eq. (3.19)] for each cell. For instance,

$$e_{(4,1,1)} = \frac{n_4^1 \cdot n_1^2 \cdot n_1^3}{150^2} = \frac{45 \cdot 47 \cdot 50}{150 \cdot 150} = 4.7$$

The expected counts are the same for all three values of X_3 and are given in Table 3.7.

The value of the χ^2 statistic [Eq. (3.20)] is given as

$$\chi^2 = 231.06$$

Using Eq. (3.21), the number of degrees of freedom is given as

$$q = 4 \cdot 3 \cdot 3 - (4+3+3) + 2 = 36 - 10 + 2 = 28$$

In Figure 3.4 the counts in bold are the dependent parameters. All other counts are independent. In fact, any eight distinct cells could have been chosen as the dependent parameters.

For a significance level of $\alpha = 0.01$, the critical value of the chi-square distribution is $z = 48.28$. The observed value of $\chi^2 = 231.06$ is much greater than z, and it is thus extremely unlikely to happen under the null hypothesis. We conclude that the three attributes are not 3-way independent, but rather there is some dependence between them. However, this example also highlights one of the pitfalls of multiway contingency analysis. We can observe in Figure 3.4 that many of the observed counts are zero. This is due to the fact that the sample size is small, and we cannot reliably estimate all the multiway counts. Consequently, the dependence test may not be reliable as well.

3.4 DISTANCE AND ANGLE

With the modeling of categorical attributes as multivariate Bernoulli variables, it is possible to compute the distance or the angle between any two points $\mathbf{x}_i$ and $\mathbf{x}_j$:

$$\mathbf{x}_i = \begin{pmatrix} \mathbf{e}_{1i_1} \\ \vdots \\ \mathbf{e}_{d\,i_d} \end{pmatrix} \qquad \mathbf{x}_j = \begin{pmatrix} \mathbf{e}_{1j_1} \\ \vdots \\ \mathbf{e}_{d\,j_d} \end{pmatrix}$$

The different measures of distance and similarity rely on the number of matching and mismatching values (or symbols) across the d attributes $\mathbf{X}_k$. For instance, we can compute the number of matching values s via the dot product:

$$s = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^{d} (\mathbf{e}_{ki_k})^T \mathbf{e}_{kj_k}$$

On the other hand, the number of mismatches is simply $d - s$. Also useful is the norm of each point:

$$\|\mathbf{x}_i\|^2 = \mathbf{x}_i^T \mathbf{x}_i = d$$

Euclidean Distance

The Euclidean distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ is given as

$$\delta(\mathbf{x}_i, \mathbf{x}_j) = \left\| \mathbf{x}_i - \mathbf{x}_j \right\| = \sqrt{\mathbf{x}_i^T \mathbf{x}_i - 2\mathbf{x}_i \mathbf{x}_j + \mathbf{x}_j^T \mathbf{x}_j} = \sqrt{2(d-s)}$$

Thus, the maximum Euclidean distance between any two points is $\sqrt{2d}$, which happens when there are no common symbols between them, that is, when $s = 0$.

Hamming Distance

The *Hamming distance* between $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined as the number of mismatched values:

$$\delta_H(\mathbf{x}_i, \mathbf{x}_j) = d - s = \frac{1}{2}\delta(\mathbf{x}_i, \mathbf{x}_j)^2$$

Hamming distance is thus equivalent to half the squared Euclidean distance.

Cosine Similarity

The cosine of the angle between $\mathbf{x}_i$ and $\mathbf{x}_j$ is given as

$$\cos\theta = \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\| \cdot \|\mathbf{x}_j\|} = \frac{s}{d}$$

Jaccard Coefficient

The *Jaccard Coefficient* is a commonly used similarity measure between two categorical points. It is defined as the ratio of the number of matching values to the number of distinct values that appear in both $\mathbf{x}_i$ and $\mathbf{x}_j$, across the d attributes:

$$J(\mathbf{x}_i, \mathbf{x}_j) = \frac{s}{2(d-s)+s} = \frac{s}{2d-s}$$

where we utilize the observation that when the two points do not match for dimension k, they contribute 2 to the distinct symbol count; otherwise, if they match, the number of distinct symbols increases by 1. Over the $d-s$ mismatches and s matches, the number of distinct symbols is $2(d-s)+s$.

Example 3.13. Consider the 3-dimensional categorical data from Example 3.11. The symbolic point (`Short`, `Medium`, `iris-versicolor`) is modeled as the vector

$$\mathbf{x}_1 = \begin{pmatrix} \mathbf{e}_{12} \\ \mathbf{e}_{22} \\ \mathbf{e}_{31} \end{pmatrix} = (0,1,0,0 \mid 0,1,0 \mid 1,0,0)^T \in \mathbb{R}^{10}$$

and the symbolic point (`VeryShort`, `Medium`, `iris-setosa`) is modeled as

$$\mathbf{x}_2 = \begin{pmatrix} \mathbf{e}_{11} \\ \mathbf{e}_{22} \\ \mathbf{e}_{32} \end{pmatrix} = (1,0,0,0 \mid 0,1,0 \mid 0,1,0)^T \in \mathbb{R}^{10}$$

The number of matching symbols is given as

$$\begin{aligned} s = \mathbf{x}_1^T \mathbf{x}_2 &= (\mathbf{e}_{12})^T \mathbf{e}_{11} + (\mathbf{e}_{22})^T \mathbf{e}_{22} + (\mathbf{e}_{31})^T \mathbf{e}_{32} \\ &= \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ &= 0 + 1 + 0 = 1 \end{aligned}$$

The Euclidean and Hamming distances are given as

$$\delta(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{2(d-s)} = \sqrt{2 \cdot 2} = \sqrt{4} = 2$$
$$\delta_H(\mathbf{x}_1, \mathbf{x}_2) = d - s = 3 - 1 = 2$$

The cosine and Jaccard similarity are given as

$$\cos\theta = \frac{s}{d} = \frac{1}{3} = 0.333$$
$$J(\mathbf{x}_1, \mathbf{x}_2) = \frac{s}{2d-s} = \frac{1}{5} = 0.2$$

3.5 DISCRETIZATION

Discretization, also called *binning*, converts numeric attributes into categorical ones. It is usually applied for data mining methods that cannot handle numeric attributes. It can also help in reducing the number of values for an attribute, especially if there is noise in the numeric measurements; discretization allows one to ignore small and irrelevant differences in the values.

Formally, given a numeric attribute X, and a random sample $\{x_i\}_{i=1}^n$ of size n drawn from X, the discretization task is to divide the value range of X into k consecutive intervals, also called *bins*, by finding $k-1$ boundary values $v_1, v_2, \dots, v_{k-1}$ that yield the k intervals:

$$[x_{\min}, v_1],\ (v_1, v_2],\ \dots,\ (v_{k-1}, x_{\max}]$$

where the extremes of the range of X are given as

$$x_{\min} = \min_i\{x_i\} \qquad\qquad x_{\max} = \max_i\{x_i\}$$

The resulting k intervals or bins, which span the entire range of X, are usually mapped to symbolic values that comprise the domain for the new categorical attribute X.

Equal-Width Intervals

The simplest binning approach is to partition the range of X into k *equal-width* intervals. The interval width is simply the range of X divided by k:

$$w = \frac{x_{\max} - x_{\min}}{k}$$

Thus, the ith interval boundary is given as

$$v_i = x_{\min} + iw, \text{ for } i = 1, \dots, k-1$$

Equal-Frequency Intervals

In *equal-frequency* binning we divide the range of X into intervals that contain (approximately) equal number of points; equal frequency may not be possible due to repeated values. The intervals can be computed from the empirical quantile or inverse cumulative distribution function $\hat{F}^{-1}(q)$ for X [Eq. (2.2)]. Recall that $\hat{F}^{-1}(q) = \min\{x \mid P(X \le x) \ge q\}$, for $q \in [0,1]$. In particular, we require that each interval contain $1/k$ of the probability mass; therefore, the interval boundaries are given as follows:

$$v_i = \hat{F}^{-1}(i/k) \text{ for } i = 1, \ldots, k-1$$

Example 3.14. Consider the `sepal length` attribute in the Iris dataset. Its minimum and maximum values are

$$x_{\min} = 4.3 \qquad\qquad x_{\max} = 7.9$$

We discretize it into $k = 4$ bins using equal-width binning. The width of an interval is given as

$$w = \frac{7.9 - 4.3}{4} = \frac{3.6}{4} = 0.9$$

and therefore the interval boundaries are

$$v_1 = 4.3 + 0.9 = 5.2 \qquad v_2 = 4.3 + 2 \cdot 0.9 = 6.1 \qquad v_3 = 4.3 + 3 \cdot 0.9 = 7.0$$

The four resulting bins for `sepal length` are shown in Table 3.1, which also shows the number of points n_i in each bin, which are not balanced among the bins.

For equal-frequency discretization, consider the empirical inverse cumulative distribution function (CDF) for `sepal length` shown in Figure 3.5. With $k = 4$ bins, the bin boundaries are the quartile values (which are shown as dashed lines):

$$v_1 = \hat{F}^{-1}(0.25) = 5.1 \qquad v_2 = \hat{F}^{-1}(0.50) = 5.8 \qquad v_3 = \hat{F}^{-1}(0.75) = 6.4$$

The resulting intervals are shown in Table 3.8. We can see that although the interval widths vary, they contain a more balanced number of points. We do not get identical

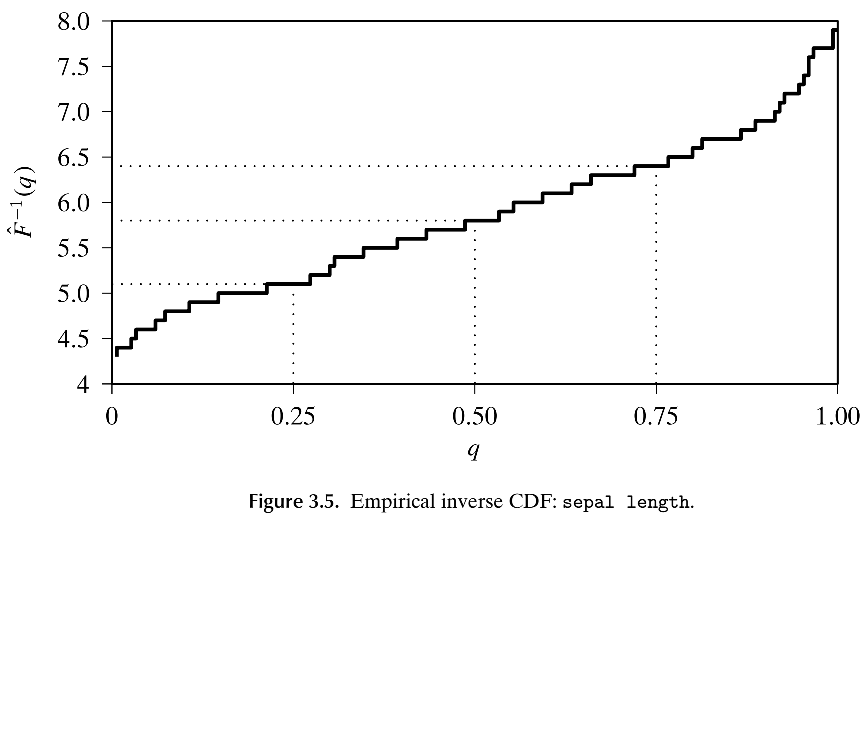

Figure 3.5. Empirical inverse CDF: `sepal length`.

Table 3.8. Equal-frequency discretization: `sepal length`

Bin	Width	Count
$[4.3, 5.1]$	0.8	$n_1 = 41$
$(5.1, 5.8]$	0.7	$n_2 = 39$
$(5.8, 6.4]$	0.6	$n_3 = 35$
$(6.4, 7.9]$	1.5	$n_4 = 35$

counts for all the bins because many values are repeated; for instance, there are nine points with value 5.1 and there are seven points with value 5.8.

3.6 FURTHER READING

For a comprehensive introduction to categorical data analysis see Agresti (2012). Some aspects also appear in Wasserman (2004). For an entropy-based supervised discretization method that takes the class attribute into account see Fayyad and Irani (1993).

Agresti, A. (2012). *Categorical Data Analysis*. 3rd ed. Hoboken, NJ: John Wiley & Sons.

Fayyad, U. M. and Irani, K. B. (1993). Multi-interval Discretization of Continuous-valued Attributes for Classification Learning. *Proceedings of the 13th International Joint Conference on Artificial Intelligence*. Morgan-Kaufmann, pp. 1022–1027.

Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. New York: Springer Science + Business Media.

3.7 EXERCISES

Q1. Show that for categorical points, the cosine similarity between any two vectors in lies in the range $\cos\theta \in [0, 1]$, and consequently $\theta \in [0^\circ, 90^\circ]$.

Q2. Prove that $E[(\mathbf{X}_1 - \boldsymbol{\mu}_1)(\mathbf{X}_2 - \boldsymbol{\mu}_2)^T] = E[\mathbf{X}_1\mathbf{X}_2^T] - E[\mathbf{X}_1]E[\mathbf{X}_2]^T$.

Table 3.9. Contingency table for Q3

	$Z = f$		$Z = g$	
	$Y = d$	$Y = e$	$Y = d$	$Y = e$
$X = a$	5	10	10	5
$X = b$	15	5	5	20
$X = c$	20	10	25	10

Table 3.10. χ^2 Critical values for different *p-values* for different degrees of freedom (q): For example, for $q = 5$ degrees of freedom, the critical value of $\chi^2 = 11.070$ has *p-value* $= 0.05$.

q	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548

Q3. Consider the 3-way contingency table for attributes X, Y, Z shown in Table 3.9. Compute the χ^2 metric for the correlation between Y and Z. Are they dependent or independent at the 95% confidence level? See Table 3.10 for χ^2 values.

Q4. Consider the "mixed" data given in Table 3.11. Here X_1 is a numeric attribute and X_2 is a categorical one. Assume that the domain of X_2 is given as $dom(X_2) = \{a, b\}$. Answer the following questions.

(a) What is the mean vector for this dataset?

(b) What is the covariance matrix?

Q5. In Table 3.11, assuming that X_1 is discretized into three bins, as follows:

$$c_1 = (-2, -0.5]$$
$$c_2 = (-0.5, 0.5]$$
$$c_3 = (0.5, 2]$$

Answer the following questions:

(a) Construct the contingency table between the discretized X_1 and X_2 attributes. Include the marginal counts.

(b) Compute the χ^2 statistic between them.

(c) Determine whether they are dependent or not at the 5% significance level. Use the χ^2 critical values from Table 3.10.

Table 3.11. Dataset for Q4 and Q5

X_1	X_2
0.3	a
−0.3	b
0.44	a
−0.60	a
0.40	a
1.20	b
−0.12	a
−1.60	b
1.60	b
−1.32	a